

# Paper 9 Technical Supplement

## Dependency-Closed Conditional Package for Response-Induced Geometry of the Admissibility Substrate

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### Abstract

This Technical Supplement is a dependency-closed conditional proof package for Paper 9, *The Geometric Substrate as Cost Structure of Comparison Continuations*. It reproduces the premises used by the certified native chain and states every standard mathematical import with its exact hypotheses. The finite operational-basis theorem is proved in conditional general form and machine-certified on the finite atom-cover instantiation; a separate physical realization gate is required before an operational basis becomes a precision-channel basis. The record topology, differential-space construction, subcartesian locus, continuation-orbit geometry, derivative-kernel identity, and response-orbit quotient gauge are each stated with their named channel, countability, smooth-generation, flow, and first-order-faithfulness gates. A physical comparison-substrate anchor, quadraticity, Lorentzian signature, the weak-field no-trace source law, and the nonlinear Paper 6 endpoint remain open, conditional, or external. No external APF reading is required for the certified native core; Paper 6 remains the separately marked source of record for the represented nonlinear endpoint.

### Governing claim.

Geometry is the represented intrinsic structure of held-profile response, after contextual identity, null response directions, record formation, ledger calibration, substrate anchoring, quadraticity, and causal signature are kept type-distinct.

$$\begin{aligned}
 \mathcal{A}_{\text{fin}} &\longrightarrow (P^{\text{red}}, \tau_{\text{rec}}, \mathcal{C}_{\text{APF}}) \\
 &\longrightarrow \text{finite differential presentation} \\
 &\longrightarrow \text{continuation orbits} \longrightarrow TS/N^{\text{ctx}} \\
 &\longrightarrow F_{\text{resp}} \longrightarrow d_{\text{resp}} \longrightarrow \text{calibrated cost geometry} \\
 &\longrightarrow \text{signed causal geometry} \longrightarrow \text{volume response.}
 \end{aligned}$$

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# 1. Closure certificate and provenance ledger

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|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Package                       | Paper 9 Technical Supplement, Version 1.1                                                                                                                                                                                                                                                                                                                                                                                                         |
| Canonical main paper          | Paper 9 — <i>The Geometric Substrate as Cost Structure of Comparison Continuations</i>                                                                                                                                                                                                                                                                                                                                                            |
| Upstream material reproduced  | Paper 1 distinction/identity closure; Paper 10 continuation calculus and the conditional finite operational-basis package; Paper 14 held continuation profile; Paper 37 held-to-record typing reproduced as a declared structural reading                                                                                                                                                                                                         |
| Standard imports stated       | Initial-topology separation; Sikorski differential spaces; subcartesian realization; complete-family orbit theorem; constant-rank quotient theorem                                                                                                                                                                                                                                                                                                |
| Certified native scope        | Conditional construction from jointly realizable operational classes to record topology, differential space, regular continuation orbits, derivative-faithful response quotient, and intrinsic response-orbit gauge                                                                                                                                                                                                                               |
| Required external APF reading | None for the certified native core; Paper 6 is required only to audit the separately marked represented nonlinear GR endpoint                                                                                                                                                                                                                                                                                                                     |
| Named conditional/open gates  | Finite joint extension and compatible joint minimum realization in the general physical domain; operational-basis-to-precision-channel realization; countable topology generation; finite smooth generation; complete local continuation flows; smooth response-factor completeness; physical substrate anchor; ledger calibration; quadraticity or Finsler branch; causal signature; response Hessian/source; no-trace; nonlinear GR translation |
| Engine status                 | APF-Engine v24.3.427 contains the exact finite atom-cover basis triad and the split Interface Engine exports/obstructions; publication freeze additionally requires a clean full-bank run, reviewed atlas re-pin, and regenerated reviewer artifacts                                                                                                                                                                                              |
| Certificate status            | Dependency-closed conditional package; physical-realization closure remains explicitly open                                                                                                                                                                                                                                                                                                                                                       |

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**Provenance rule.** A theorem may have a canonical home elsewhere in the APF corpus while being reproduced here in full. Canonical ownership controls maintenance; local reproduction controls reviewer self-containment. The local statements below govern this supplement.

## 2. Closed upstream operational foundation

**Definition 2.1** (Enforceable distinction). *An enforceable distinction on a physical interface  $U$  is a tested alternative whose two outcomes can be maintained, propagated through an admissible continuation, and deposited in a physical record with nonzero operational consequence. Two descriptions that differ only by data with no possible continuation or record witness represent the same physical distinction class.*

**Definition 2.2** (Closed finite interface and marginal floor). *A closed interface  $U$  has finite available enforcement capacity  $C_U < \infty$ . There is a positive marginal distinction floor  $\varepsilon_* > 0$ : every genuinely*

additional tested distinction committed jointly with an existing realization increases minimum committed capacity by at least  $\varepsilon_*$ .

**Principle 2.3** (Contextual continuation identity). *Physical identity is equality across every admissible future continuation context. Consequently, a coordinate omitted from a purportedly complete carrier is physically consequential only when some admissible continuation distinguishes the omission; a retained coordinate with no such witness is quotient-null.*

**Principle 2.4** (Closed-world completeness). *Every enforceable distinction of a closed interface belongs to its complete physical ledger. Conversely, a nonzero element of the reduced physical representation must be exposed by at least one admitted comparison or record sector. Hidden consequential distinctions and physically empty represented directions are both forbidden in the reduced closed-world description.*

**Principle 2.5** (A2 no-excess realization). *For fixed enforceable distinction content, the realized carrier is a minimum-cost admissible realization. A carrier may not retain positive-cost capacity that contributes no additional enforceable distinction. This no-excess clause compares nested distinction content; it does not choose among physically distinct outcomes.*

**Definition 2.6** (Held continuation profile). *A held continuation profile is a contextual equivalence class*

$$H = [\mathcal{P}]_{\text{ctx}}$$

*of pre-record physical presentations that agree in every admissible future continuation context. Particular domains, groups, coordinates, or amplitudes are presentations of  $H$ , not the ontology of  $H$  itself.*

**Definition 2.7** (Record quotient and held-to-record transition). *The invariant record quotient forgets continuation data that no committed readout can retain. An outcome-bearing record transition is a distinct map from a held profile to a committed record sector. Thus the held carrier, the invariant record quotient, and the outcome-selection instrument are not identified.*

*Audit Note 2.8* (Typing status). The three-way distinction above is a declared structural typing, reproduced at the same reading grade as its upstream source. It is not promoted here to a primitive-level theorem that physically derives the held carrier, quotient, or outcome instrument. Equality of concrete presentations is never substituted for canonical quotient or isomorphism.

**Definition 2.9** (Operational recoverability and closure). *For operational realizations  $a, b$  on  $U$ , write  $a \preceq_U b$  when every tested distinction carried by  $a$  is recoverable from  $b$  by admissible deterministic lossless postprocessing. Mutual recoverability defines classes  $[a]_U$ . For a family  $S$ ,  $\text{cl}_U(S)$  is the set of classes recoverable from a joint realization of  $S$ .*

**Definition 2.10** (Joint-realizability package). *Let  $\mathcal{A}_U$  be the admitted operational classes and let  $\mathfrak{J}_U \subseteq \mathcal{P}_{\text{fin}}(\mathcal{A}_U)$  be the downward-closed family of jointly realizable finite class sets. For each  $S \in \mathfrak{J}_U$ , a no-excess minimum joint realization exists and has cost*

$$\kappa_U(S) := \min_{r \in \mathcal{R}_U(S)} \kappa_U(r).$$

*All such minima have the same deterministic-lossless recoverability closure, which defines  $\text{cl}_U(S)$ .*

**Assumption 2.11** (Finite joint extension / amalgamation). *If  $S \in \mathfrak{J}_U$  and  $\text{cl}_U(S) \neq \mathcal{A}_U$ , then there exists  $[a]_U \notin \text{cl}_U(S)$  for which  $S \cup \{[a]_U\} \in \mathfrak{J}_U$ .*

**Definition 2.12** (Positive marginal extension floor). *For every jointly realizable new-class extension,*

$$\kappa_U(S \cup \{[a]_U\}) - \kappa_U(S) \geq \varepsilon_* > 0.$$

*This is the typed A2 no-excess plus positive tested-distinction-floor statement used below.*

**Lemma 2.13** (A2 distinction-increment lemma — typed). *If  $S, S \cup \{[a]_U\} \in \mathfrak{J}_U$  and  $[a]_U \notin \text{cl}_U(S)$ , then*

$$\kappa_U(S \cup \{[a]_U\}) \geq \kappa_U(S) + \varepsilon_*.$$

*Proof.* Immediate from Definition 2.12. □

**Theorem 2.14** (Finite Operational Basis Theorem — conditional general form). *Let  $U$  satisfy finite capacity, the joint-realizability package of Definition 2.10, finite joint extension (Assumption 2.11), the positive marginal extension floor, and closure under deterministic lossless postprocessing. Then there exists  $B_U \in \mathfrak{J}_U$  with*

$$\text{cl}_U(B_U) = \mathcal{A}_U, \quad |B_U| \leq \left\lfloor \frac{C_U}{\varepsilon_*} \right\rfloor.$$

*A closure-redundant member may be deleted until  $B_U$  is inclusion-minimal.*

*Proof.* Starting from  $S_0 = \emptyset$ , use finite joint extension whenever closure is incomplete. The typed increment lemma gives  $\kappa_U(S_k) \geq k\varepsilon_*$ , while joint admissibility gives  $\kappa_U(S_k) \leq C_U$ . Hence the process stops after at most  $\lfloor C_U/\varepsilon_* \rfloor$  extensions. Downward closure of  $\mathfrak{J}_U$  permits redundant-member deletion. □

**Corollary 2.15** (Finite compatible minimum joint realization). *For a complete basis  $B_U = \{[a_1]_U, \dots, [a_N]_U\}$ , minimum attainment gives one no-excess joint carrier  $r_{B_U}$  and deterministic lossless projections  $\pi_i(r_{B_U}) \in [a_i]_U$ . The projected family is finite and compatible because all members descend from the same joint carrier. Independently chosen classwise minima are not asserted to be compatible.*

*Audit Note 2.16* (Finite atom-cover Engine instantiation). At APF-Engine v24.3.427 (commit 7ab898e5ad48aefa620e07e928771727236a8b20) or its reviewed integration successor, the three checks

```
T_finite_operational_basis_scope_contract, T_finite_operational_basis,
T_finite_minimal_joint_realization_atom_cover
```

certify the exact finite atom-cover instantiation: finite admitted atoms, subset-coded classes, union joint carrier and closure, exact no-excess atom cost, and one compatible minimum joint carrier. They do not derive finite joint extension for an arbitrary physical operational closure.

*Boundary 2.17* (Scope). Finite operational generation does not imply finitely many terminal values, a finite state space, a manifold, or finite smooth generation. Precision-refinement channels and differential presentation are separate downstream questions.

### 3. Standard mathematical imports

**Theorem 3.1** (Initial-topology separation import). *A family of real-valued functions separating points generates a Hausdorff initial topology. If a countable subfamily generates that topology and rational intervals are used as a base in each codomain, the resulting topology is second countable.*

**Theorem 3.2** (Sikorski differential-space import). *Let  $P$  be a topological space and  $\mathcal{C}$  a family of real-valued functions closed under localization and smooth superposition: for  $f_1, \dots, f_m \in \mathcal{C}$  and  $\psi \in C^\infty(\mathbb{R}^m)$ ,  $\psi(f_1, \dots, f_m) \in \mathcal{C}$ . Then  $(P, \mathcal{C})$  is a differential space. Tangent vectors at  $p$  may be defined as derivations of the local algebra.*

**Theorem 3.3** (Subcartesian realization import). *If every point of a Hausdorff differential space has a neighbourhood differentially isomorphic to a differential subspace of some  $\mathbb{R}^N$ , the space is subcartesian. A finite map  $F = (f_1, \dots, f_N)$  supplies such a realization when it is a homeomorphism onto its image and every local smooth observable factors through  $F$  by a smooth Euclidean function.*

**Theorem 3.4** (Complete-family orbit import). *On a subcartesian space, the orbit of a family of vector fields that admits local flows and is complete under pushforward by those flows is an immersed smooth manifold. Its tangent space is the span of the vector-field values along the orbit.*

**Theorem 3.5** (Constant-rank quotient import). *For a smooth bundle map of locally constant rank, the kernel is a smooth subbundle, the quotient by the kernel is a smooth vector bundle, and the induced map from the quotient to the image is a bundle monomorphism.*

## 4. Dependency spine of the conditional package

The proof DAG used in this supplement is

finite capacity + joint-realizability package  
 + finite joint extension + positive marginal extension floor  
 $\implies$  finite operational basis (conditional general; exact atom-cover witness)  
 $\xrightarrow{\text{faithful operational-to-channel gate}}$  finite separating precision-channel basis  
 $\xrightarrow{\text{quotient-faithful real limits + topology generation}}$  Hausdorff, second-countable record topology  
 $\xrightarrow{\text{finite smooth generation}}$  subcartesian APF differential space  
 $\xrightarrow{\text{complete local continuation flows}}$  smooth continuation orbits  
 $\xrightarrow{\text{smooth response-factor completeness}}$   $N^{\text{ctx}} = \ker D\beta$   
 $\implies$  intrinsic response-orbit quotient gauge  
 $\xrightarrow{\text{physical substrate anchor}}$  physical comparison-substrate geometry.

Every arrow carrying a label is a named gate rather than a consequence of the preceding line alone. The causal-signature, weak-field no-trace, and nonlinear Paper 6 translation gates occur strictly downstream.

## 5. Orientation: the geometric-substrate problem

Paper 10 supplied a continuation-first language for APF. Its central move was to separate admissible possibility from dynamics:

$$\Gamma \Vdash_C D \rightsquigarrow E$$

states that  $D$  can continue as  $E$  within capacity bound  $C$ . A dynamics, probability law, or variational rule is a later selector or weighting on admissible continuation words.

The same paper proposed a compressed reading of geometry:

Geometry is the regularized cost structure of carrying comparison, localization, and transport continuations.

This paper begins the work required to make that sentence theorem-grade.

The geometric-substrate problem has six type-distinct parts:

1. define comparison continuations and finite response models operationally;
2. quotient response-null coordinates and construct intrinsic response length;
3. calibrate that modeled length to the physical interaction ledger;
4. anchor response directions to tangent directions of the comparison substrate;
5. derive a signed causal comparison geometry without confusing it with the positive cost metric;
6. determine how recruitment load deforms the comparison geometry and align the result with the Paper 6 represented-GR closure.

This revision proves the finite geometric statements in items 2 and the conditional calibration statement in item 3. It formulates item 4 as an explicit anchor gate, corrects the signature logic in item 5, and retains the redshift and scalar trace program as the first worked application of item 6. *Boundary* 5.1 (Status discipline). Throughout this paper, P means proved from stated definitions and assumptions in this paper; C means conditional on an explicit gate or structural commitment; R means recovered in a represented regime; I means imported from another APF paper under its stated hypotheses; and O means open. The present document mixes P, C, R, and I: the quotient and pullback theorems are finite mathematics; ledger realization, anchoring, signature, and the no-trace source are conditional or open; redshift is a conditional recovery; and the Einstein closure is imported from Paper 6.

## On operational language

This paper develops geometry as the represented cost structure of finite comparison continuations. The continuation primitive imported from Paper 10 is a static algebraic relation  $\Gamma \Vdash_C D \rightsquigarrow E$  on continuation profiles, and the metric-emergence ladder of §12 derives the quadratic metric kernel as a representation *conditional on a series of structural gates closing simultaneously* — not as something the framework’s primitives produce by temporal evolution.

The phrasings “becomes available,” “emerges,” “closes,” and “recovers” that appear throughout this paper are the local readings of conditional structural commitments. Where the prose says “the metric becomes available only after several gates close,” the structural reading is “the metric is a representation conditional on the gates’ joint structural validity.” Where the prose says “admissible chart changes become diffeomorphism-type relabelings in the smooth represented regime,” the structural reading is “in the smooth represented regime, the admissible relabelings are exactly the diffeomorphism-type ones.” The “becomes” is locator-language, not process-language.

Time itself is derived in this corpus: Paper 3 derives the arrow of time as the loss of global admissible coordination, and Paper 6 derives the metric and dimensionality of spacetime. The geometric machinery developed here operates on the static comparison structure  $(c_x, q_x)$  at the substrate; what reads as “response,” “recruitment,” or “redshift recovery” is the local reading of conditional structural commitments under temporal slicing. A reader who wants the structural reading can take every “becomes available,” “emerges,” and “recovers” as shorthand for the local reading of the corresponding structural conditional. The technical apparatus developed below — the cost germ  $c_x$ , the quadratic kernel  $q_x$ , the geometric recruitment functional, the cost-curvature response — is already structural; only the pedagogical wrapping is temporal.

This convention matches Paper 0’s “A note on language” in the Prelude §1 and Paper 0’s Descriptive



Reading chapter on temporal language and the eternalist commitment, Paper 1 supplement v8.24’s “On the use of operational language” note in §1, and Paper 10 v1.17’s parallel note in §1. The framework is in this respect in the company of block-universe readings of general relativity, the Wheeler–DeWitt timeless quantum gravity programme, and the Page–Wootters mechanism for time emergence in entangled quantum systems — structural at the foundation, with time as a derived feature of how the structure is read locally.

*A note on two “substrate” senses.* The framework uses “substrate” in two distinct but related senses, and a reader of this paper alongside Paper 0 should keep them apart. Paper 9 uses *substrate* in its **material-substrate** sense throughout: the *geometric substrate* is the physical object that carries comparison continuations; the *comparison substrate* is the physical object equipped with admissible comparison continuations between sites; *substrate degrees of freedom* are the local mode-content that the recruitment profile  $\varphi_d(x)$  recruits to carry a continuation. Paper 0 §sec:substrate\_position\_axis introduces a different sense: the *substrate-position axis* is a position locus on the framework’s three-regime architecture (void regime / middle regime / saturation regime). The two senses are parallel but not identical: the geometric substrate is *what carries* the comparison continuation; the substrate-position axis is *where in regime space* the architecture sits. Paper 9’s content is largely middle-regime physics in the substrate-position-axis sense (the metric-emergence ladder of §12 requires middle-regime conditions: locally smooth and locally additive comparison cost on a connected comparison substrate without saturation); the saturation-direction reading is articulated in §“Cost-curvature response” where recruitment-load already committed reduces residual stiffness for additional comparison continuations. Paper 6 main v3.9 + Supplement v2.10 carry the substrate-position-axis architectural reading at the dynamics/geometry/cosmology level.

## 6. Canonical type ledger and corpus dependencies

Paper 9 consumes the mature continuation architecture rather than redefining it. The canonical objects are:

| Object                   | Meaning                                                            | Source of record                                                |
|--------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|
| $H \in \text{Held}$      | Structurally aligned, occupied, pre-record continuation profile    | Papers 1, 10, and 14                                            |
| $V_H$                    | Complete finite represented carrier of held continuation profiles  | Papers 1 and 10; Paper 5 supplies the quantum-sector completion |
| $\mathfrak{T}_H$         | Represented algebra of admissible transformations of held profiles | Papers 1 and 5                                                  |
| $\mathcal{A}_H$          | Observable/effect algebra paired with held states and processes    | Paper 5                                                         |
| $R(H) \in \text{InvRec}$ | Invariant phase-insensitive committed-record image                 | Papers 10 and 37                                                |
| $\{C_i(H)\}$             | Outcome-bearing committed records                                  | Papers 5 and 37                                                 |
| $R_\Pi / \sim$           | Response coordinates modulo process-null implementation directions | Paper 9                                                         |

The geometry pipeline of this paper is therefore

$$\begin{aligned} H \in \mathbf{Held} &\longrightarrow V_H \longrightarrow \mathfrak{T}_H, \\ T\mathcal{M} &\longrightarrow R_\Pi / \sim \longrightarrow \mathfrak{T}_H. \end{aligned}$$

with the final arrow supplied only by a physical comparison-response anchor. Record formation  $Q : \mathbf{Held} \rightarrow \mathbf{Rec}$  is downstream and is not used to define the pre-record response geometry.

*Audit Note 6.1* (Three distinct planes and three distinct orientations). The elementary quantum comparison carrier of Paper 5, the clock-ruler load-response plane of this paper, and the tangent plane of a Lorentzian comparison substrate are not automatically identical:

$$W_A^{\text{quantum}} \neq W_e^{\text{clock-ruler}} \neq T_x \mathcal{M}$$

unless an explicit anchor theorem identifies them. Likewise distinguish

$$\begin{aligned} o_H &= \text{held complex orientation,} \\ o_{\text{rec}} &= \text{irreversible record order,} \\ o_{\text{caus}} &= \text{time orientation of the causal cone.} \end{aligned}$$

Paper 5 derives  $o_H$  in its declared coherent quantum regime. Papers 3 and 37 supply the irreversible record order  $o_{\text{rec}}$ . Paper 9 may represent that order by  $o_{\text{caus}}$  only after the causal-cone and signed-signature gates close. No one of these orientations is substituted for another by notation.

*Audit Note 6.2* (Cross-paper dependency boundary). Paper 10 supplies continuation grammar, held-profile identity, and the universal held continuation lift. Paper 5 supplies the quantum-sector represented carrier, local coherent circle, and natural complex orientation. Paper 9 neither rederives those quantum results nor uses them to close the substrate anchor, the Lorentzian signature gate, or the weak-field no-trace operator. Paper 6 remains the source of record for the represented nonlinear Einstein closure.

## 7. Main results and non-claims of this draft

The paper now has two linked theorem spines. The first is intrinsic response geometry; the second is the weak-field deformation of the represented comparison kernel. The table records what is proved, what is conditional, and what remains open.

| Result                   | Status      | Meaning                                                                                                                                                                                                                                                                 |
|--------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite operational basis | Cond./model | The general theorem is derived only after finite joint extension, minimum attainment, closure invariance, and the positive marginal extension floor are named. The Engine certifies the exact finite atom-cover instantiation and one compatible minimum joint carrier. |

|                                              |     |                                                                                                                                                                                                                                          |
|----------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite physical precision-channel generation | G   | Requires the faithful operational-to-channel realization gate; the operational basis theorem alone does not supply channel towers, state separation, or real limits.                                                                     |
| Record topology                              | P G | Quotient-faithful real channel limits generate the initial topology; a separating family gives Hausdorffness. Second countability additionally requires a countable specification grammar and a countable topology-generating subfamily. |
| Differential-space closure                   | P/D | Lossless deterministic postprocessing preserves physical fibres; localization and smooth saturation define the minimal Sikorski differential structure compatible with the admitted readout algebra.                                     |
| Finite-channel sufficiency                   | P/G | Once a complete jointly realizable channel basis exists, additive standing-capacity accounting makes it finite. Existence is supplied only conditionally by the operational-basis theorem plus the faithful realization gate.            |
| Finite differential sufficiency              | G   | After finite state separation, one must still prove finite smooth generation of the continuation-regular sheaf. This is the subcartesian regularity gate.                                                                                |
| Smooth continuation orbits                   | I/P | On the finite-differential-presentation locus, the imported subcartesian orbit theorem makes complete-family continuation orbits immersed smooth manifolds.                                                                              |
| Regular local response quotient              | P H | Constant rank gives local quotient charts. The derivative-kernel identity further requires smooth two-way factorization between the complete response and a finite local APF generator; equal fibres alone are insufficient.             |
| Response-orbit quotient gauge                | P H | After the smooth-factor gate closes, the injective quotient response pulls the process norm back to a positive homogeneous gauge on the response orbit.                                                                                  |
| Physical comparison-substrate anchor         | G   | Requires a site map and a tangent map equal to the quotient differential of that site map, with direct-comparison calibration and stability.                                                                                             |
| Intrinsic response distance                  | P   | Integrating the response-orbit gauge gives a coordinate-invariant path distance that dominates endpoint process displacement.                                                                                                            |
| Linear Paper 2 specialization                | P   | For $\Phi(\mathcal{F}, r) = \mathcal{F} + T_{\Pi}r$ , the geometry is the flat quotient $R_{\Pi}/\ker T_{\Pi}$ and reproduces the optimal response gains.                                                                                |
| Physical-ledger calibration                  | P H | Pointwise comparison of ledger rate with the pull-back gauge gives bi-Lipschitz comparison; exact equality requires exact calibration.                                                                                                   |

|                                    |       |                                                                                                                                                     |
|------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Local residual energy–distance law | P SRR | A smooth positive Hessian gives $I_{\Pi}^{\text{res}} = \frac{1}{2}d_{\mathcal{G}_E}^2 + o(d^2)$ near equilibrium.                                  |
| Quadratic positive cost metric     | P Q   | A parallelogram certificate converts the response gauge into a positive metric. Weighted $\ell_1$ ledgers remain Finsler.                           |
| Lorentzian comparison structure    | O/C   | A separate finite-coordination cone and signed two-homogeneous gate is required; no positive pullback metric can create Lorentzian signature.       |
| Point-source exterior profile      | C/P   | An assumed isotropic quadratic exterior kernel gives $\chi(r) \sim 1/r$ ; the kernel and calibration remain inputs.                                 |
| Clock-sector redshift recovery     | R     | Calibrated point-mass load, clock response, and the proper-time dictionary recover the Schwarzschild redshift factor conditionally.                 |
| Trace–volume theorem               | P/C   | For $q_{\chi} = R_{\chi}^* q_0$ , the first comparison-volume variation is $\text{tr } L$ ; after clock calibration it is $1 - \gamma_C$ .          |
| No-trace operator reduction        | P/C   | In gauge-fixed linear response the weak-field gate is $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ ; deriving it from recruitment remains open. |
| Conditional weak-field exterior    | C/R   | If no trace closes, the first-post-Newtonian Schwarzschild exterior follows in isotropic weak-field form.                                           |
| Full represented GR                | I/C   | External Paper 6 endpoint under H1–H3/A9/Lovelock; finite-continuability translation remains open.                                                  |

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*Audit Note 7.1 (Non-claims).* This paper does not identify every physical interaction cost with a distance, does not derive Lorentzian signature from  $T^*T$ , does not infer spacetime dimension from a register-count theorem, and does not turn physical divisibility into the mathematical subdivision of a curve. It does not rederive the Einstein equations, the value of  $G$ , the dimension  $(1+3)$ , or the full equivalence principle. Its new mathematical contribution is the typed record-topology–differential-space–orbit construction, the kernel-first response geometry on emergent smooth branches, and the trace–volume reduction of the weak-field spatial obstruction.

## 8. Alignment with Paper 6: represented-GR source of record

Paper 6 and its Technical Supplement already contain the represented-geometry route to GR. In that paper the sequence is:

$$\begin{aligned}
\text{cost distance} &\Rightarrow g_{\mu\nu} \Rightarrow (-, +, +, +), \quad d = 4 \\
&\Rightarrow \text{capacity-gradient curvature} \Rightarrow \text{A9 closure} \\
&\Rightarrow \text{Lovelock uniqueness} \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.
\end{aligned}$$

Paper 9 should therefore not duplicate that proof. Its role is to express the same represented-geometry closure in the newer finite-continuability language and to identify what that closure looks like in the weak-field scalar sector. Of the imported hypotheses, H1 (continuum emergence) and H2

(locality of the  $\Delta_{\text{geo}}$  closure) are discharged by the Recruitment-Radius papers (Papers 24 and 25) via continuous anchor profiles and  $L_{\text{loc}}$  on substrate kernels; the represented-geometry import that remains is H3 together with the A9/Loveloock uniqueness step.

**Theorem 8.1** (Paper 6 represented-GR import). *Under the Paper 6 hypotheses H1–H3, together with the A9 closure and Lovelock uniqueness in the represented smooth Lorentzian regime, the admissibility-preserving geometric dynamics are governed by*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

*This theorem is imported here as a represented-geometry result, not reproved.*

**Corollary 8.2** (Weak-field shadow of the Paper 6 closure). *If the Paper 6 represented-GR closure is accepted, then in the static weak-field exterior of a point mass the metric representation has the first-post-Newtonian Schwarzschild form*

$$ds^2 = - \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) d\mathbf{x}^2 + O(c^{-4}),$$

*and hence the scalar weak-field response of Paper 9 has*

$$\gamma_C = 1, \quad \theta = 1 - \gamma_C = 0.$$

*Audit Note 8.3* (What Paper 9 adds beyond importing Paper 6). The import theorem says that the represented smooth geometry satisfies the Einstein equations once the Paper 6 gates close. Paper 9 adds a lower-level translation: redshift fixes the clock response, while the entire leading spatial ambiguity is the trace of a local clock-ruler scalar response generator. Thus the no-trace/unimodular gate is the finite-continuability weak-field face of the Paper 6 A9/Loveloock closure. If Paper 6 is used,  $\theta = 0$  follows in the weak-field represented regime. If Paper 9 is developed bottom-up,  $\theta = 0$  is the next primitive substrate theorem to prove.

## 9. Reader path and theorem spine

The paper should be read in two passes. The first pass constructs intrinsic response geometry without presupposing spacetime. The second pass anchors and deforms that geometry in the weak-field comparison sector.

### Publication spine.

finite executable records  $\implies$  Hausdorff second-countable record topology  
 $\implies$  refinement-complete readout sheaf  
 $\implies$  Sikorski differential space  
 $\implies$  finite differential presentation, where certified  
 $\implies$  smooth continuation orbits  
 $\implies TS/N^{\text{ctx}} \hookrightarrow TP^{\text{red}}$   
 $\implies$  pullback gauge and calibrated physical cost  
 $\implies$  positive cost metric, if quadratic  
 $\implies$  signed causal geometry, if independently closed  
 $\implies \delta \log \sqrt{|\det q|} = \text{tr } L = 1 - \gamma_C.$

The decisive closures are therefore:

1. construct a complete regular response model for comparison continuations;
2. calibrate its tangent gauge to the physical interaction ledger;
3. construct the comparison-response anchor and the independent causal cone;
4. derive the loaded-equilibrium Hessian  $\mathcal{K}$  and source  $\mathcal{J}$ ;
5. prove the no-trace condition  $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$ .

The first two now have precise finite theorems. The third and fifth are the new APF-specific targets. Connections and curvature are deferred until regularity, smoothness, strong convexity or quadraticity, and the causal signature gate are all established.

*Audit Note 9.1* (Why Paper 9 is the geometric home). Paper 2 supplies a finite response model and its quotient estimates. Paper 9 generalizes those estimates into intrinsic path geometry, anchors them to the comparison substrate, separates positive cost geometry from signed causal geometry, and applies the result to recruitment-deformed clock-ruler response. Paper 2 should therefore import finished Paper 9 theorems later rather than absorb the geometric program itself.

## 10. Imports from finite continuability

**Definition 10.1** (Continuation judgment). *The primitive APF judgment is*

$$\Gamma \Vdash_C D \rightsquigarrow E.$$

*It means that, in context  $\Gamma$ , distinction-expression  $D$  can continue as distinction-expression  $E$  using enforcement capacity no greater than  $C$ .*

**Definition 10.2** (Continuation cost). *The cost of a continuation is*

$$\kappa_\Gamma(D \rightsquigarrow E) := \inf\{C : \Gamma \Vdash_C D \rightsquigarrow E\}.$$

*An identity-class continuation is a continuation with  $E \in [D]_\Gamma$ , where  $[D]_\Gamma$  is the tested continuation-equivalence class of  $D$ .*

**Definition 10.3** (Held continuation profile import). *A held continuation profile  $H \in \text{Held}$  is the structurally aligned, occupied, pre-record object that carries complete contextual continuation identity together with every consequential coherent distinction still available to later continuation. Its finite represented carrier is  $V_H$ . A record-forming class transition is a generally nonfaithful map*

$$Q : \text{Held} \longrightarrow \text{Rec}$$

*that commits a tested distinction and may remove globally common orientation while retaining consequential relative information.*

**Definition 10.4** (Universal held continuation lift import). *For a local comparison coordinate  $q$  inside  $H$ , Paper 10 defines the minimum complete local carrier*

$$L_q(H) = G_q(H) / \sim_q,$$

*the quotient of admissible local continuation germs by complete short-context equivalence. In a smooth operationally first-order branch it is represented by the admitted first-jet locus. Paper 9 uses this only as an identity-complete local carrier; it does not import the Paper 5 quantum  $S^1$  theorem into substrate geometry.*

**Definition 10.5** (Recruitment profile). *For a continuation  $D \rightsquigarrow E$ , the recruitment profile  $\phi_{D \rightarrow E}(x)$  records where and how strongly substrate degrees of freedom are recruited to carry that continuation. The associated recruitment cost is denoted*

$$E_{\text{rec}}[\phi; \Gamma].$$

*Fields are represented equilibrium recruitment profiles. Radiation is cost discharge from failed adiabatic recruitment. Geometry is the special case in which the recruited substrate carries comparison, localization, and transport continuations.*

*Source of the recruitment formalism.* The recruitment profile  $\phi_{D \rightarrow E}$ , the recruitment cost  $E_{\text{rec}}$ , and their dynamics are formalized in the Recruitment-Radius papers (Papers 24 and 25): the recruitment cost functional, the master equation for substrate recruitment, continuous anchor profiles, and the substrate-kernel structure. The present paper specializes that formalism to the geometric case — comparison, localization, and transport continuations. Where the derivations below call for a nonlinear recruitment functional, a substrate kernel, or its Green’s function, the source of record is Papers 24/25.

## 11. Comparison continuations and the three geometric levels

**Definition 11.1** (Comparison continuation). *A comparison continuation from  $x$  to  $y$  in context  $\Gamma$  is a substrate operation  $K : x \rightsquigarrow y$  that carries a probe’s held continuation profile from  $x$  to  $y$ , preserves its complete contextual identity class along the way, and establishes endpoint correspondence. Its ledger may be decomposed as*

$$\kappa_{\Gamma}(K) = \kappa_{\text{transport}}(K) + \kappa_{\text{identity}}(K) + \kappa_{\text{corr}}(K),$$

*with the understanding that the three terms may share physical support.*

**Definition 11.2** (Directed comparison cost and comparison substrate). *The directed comparison cost is*

$$d_{\Gamma}^{+}(x, y) := \inf\{\kappa_{\Gamma}(K) : K \text{ is an admissible comparison continuation } x \rightsquigarrow y\}.$$

*A comparison substrate  $\mathcal{M}$  is a regular domain of distinguishable comparison sites equipped with admissible nearby comparison continuations. It is locally comparison-complete at scale  $\ell$  when every represented pair below that scale is connected by at least one admissible comparison continuation.*

**Definition 11.3** (Direct local comparison germ). *Whenever the refinement limit exists, define*

$$c_x^{\text{dir}}(u) := \lim_{\epsilon \downarrow 0} \frac{d_{\Gamma}^{+}(x, x + \epsilon u)}{\epsilon}.$$

*This is the direct substrate-side cost germ. It may be asymmetric, nonsmooth, or degenerate before further gates close.*

The response construction below introduces three distinct geometries.

---

| Level | Carrier | Role |
|-------|---------|------|
|-------|---------|------|

---

|                               |                                                                                                                 |                                                                                                                                                                          |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process geometry              | Represented held-process algebra $\mathfrak{T}_H$ acting on $V_H$ , with operational distance $d_{\text{proc}}$ | Measures distinguishability or displacement between represented processes. In the Paper 2 channel model, $\ \Delta\ _{\text{proc}} = \frac{1}{2}\ \Delta\ _{\diamond}$ . |
| Response geometry             | Response-null quotient $\mathcal{Q}_{\Pi, \mathcal{F}}$                                                         | Measures intrinsic length of changing a represented response class while redundant coordinates have been removed.                                                        |
| Substrate comparison geometry | Comparison substrate $\mathcal{M}$                                                                              | Measures comparison, localization, and transport directions at physical sites after an anchor identifies them with response directions.                                  |

*Audit Note 11.4* (Type ledger). The following objects must not be silently identified:

| object                      | domain                           | homogeneity/role                         |
|-----------------------------|----------------------------------|------------------------------------------|
| $F_{\text{resp}}(q, v)$     | $T\mathcal{Q}$                   | positive one-homogeneous gauge           |
| $d_{\text{resp}}(q_0, q_1)$ | $\mathcal{Q} \times \mathcal{Q}$ | intrinsic path distance                  |
| $I_{\Pi}^{\text{path}}$     | physical protocol paths          | calibrated path ledger                   |
| $I_{\Pi}^{\text{res}}$      | endpoint response state          | local two-homogeneous residual energy    |
| $M_{\text{sep}}$            | $\mathcal{P}$                    | extrinsic distance to a free-process set |
| $Q_x^{\text{cmp}}(u)$       | $T_x\mathcal{M}$                 | signed causal comparison function.       |

A valid theorem may compare these objects, but their equality is never a consequence of notation.

## 12. Response-null quotient and intrinsic process geometry

Let  $R_{\Pi}$  be a finite-dimensional response-coordinate manifold, fix a free represented held process  $\mathcal{F} \in \mathfrak{T}_H$ , and let

$$\Phi_{\mathcal{F}} : R_{\Pi} \longrightarrow \mathfrak{T}_H, \quad \Phi_{\mathcal{F}}(0) = \mathcal{F},$$

be a differentiable response model. Coordinates can represent memory occupation, a residual correlation register, a joint repair operation, nonfactorizing control amplitude, or any other finite response variable. Coordinates are not physical merely because they are written down.

**Definition 12.1** (Response fiber and regular quotient chart). *For fixed  $\mathcal{F}$ , define*

$$r \sim_{\mathcal{F}} r' \iff \Phi_{\mathcal{F}}(r) = \Phi_{\mathcal{F}}(r').$$

*A regular local response quotient near  $r_0$  is a triple*

$$U \xrightarrow{\pi} \mathcal{Q}_{\Pi, \mathcal{F}} \xrightarrow{\bar{\Phi}_{\mathcal{F}}} \mathcal{P}$$

*such that  $\pi$  is a surjective submersion, its fibers are exactly the local response fibers,  $\Phi_{\mathcal{F}}|_U = \bar{\Phi}_{\mathcal{F}} \circ \pi$ , and  $\bar{\Phi}_{\mathcal{F}}$  is an immersion.*

**Proposition 12.2** (Local quotient from constant rank). *Suppose  $\Phi_{\mathcal{F}}$  takes values in a finite-dimensional smooth process chart, is  $C^1$ , and has constant rank  $k$  on a neighborhood of  $r_0$ . Then after shrinking the neighborhood it admits a regular local response quotient of dimension  $k$ . If the rank changes, no smooth global quotient is implied; the natural response geometry may instead be stratified or singular.*



*Proof.* The constant-rank theorem gives local coordinates  $(u, w)$  on the response domain and  $y$  on process space in which  $\Phi_{\mathcal{F}}(u, w) = (u, 0)$ . Projection  $(u, w) \mapsto u$  supplies the local quotient, and  $u \mapsto (u, 0)$  is the induced immersion. At a rank-changing point this normal form cannot be chosen with fixed quotient dimension, so smooth-manifold conclusions require an additional stratification analysis.  $\square$

**Definition 12.3** (Pullback process gauge). *Let the represented process displacement space carry a norm  $\|\cdot\|_{\text{proc}}$ . On a regular quotient define*

$$F_{\text{resp}}(q, v) := \left\| D\bar{\Phi}_{\mathcal{F}, q}(v) \right\|_{\text{proc}} \quad (q \in \mathcal{Q}_{\Pi, \mathcal{F}}, v \in T_q \mathcal{Q}_{\Pi, \mathcal{F}}).$$

For the finite channel model of Paper 2,

$$\|\Delta\|_{\text{proc}} := \frac{1}{2} \|\Delta\|_{\diamond}.$$

**Theorem 12.4** (Intrinsic pullback-gauge theorem). *On every regular quotient chart,  $F_{\text{resp}}(q, \cdot)$  is a norm on the tangent space. It is invariant under smooth reparametrization of quotient coordinates. If the ambient process norm is not smooth or strictly convex, the result is a possibly nonsmooth Finsler-type structure rather than a classical strongly convex Finsler metric.*

*Proof.* Positive homogeneity and the triangle inequality follow from the process norm and linearity of  $D\bar{\Phi}_q$ . If  $F_{\text{resp}}(q, v) = 0$ , then  $D\bar{\Phi}_q v = 0$ ; immersion of  $\bar{\Phi}$  gives  $v = 0$ . Under a quotient-coordinate change  $\psi$ , the chain rule sends  $v$  to  $D\psi(v)$  and leaves the process tangent vector  $D\bar{\Phi}(v)$  unchanged, proving invariance.  $\square$

**Definition 12.5** (Intrinsic response length and distance). *For an absolutely continuous quotient path  $\gamma : [0, 1] \rightarrow \mathcal{Q}_{\Pi, \mathcal{F}}$ , define*

$$L_{\text{resp}}[\gamma] := \int_0^1 F_{\text{resp}}(\gamma(t), \dot{\gamma}(t)) dt,$$

and

$$d_{\text{resp}}(q_0, q_1) := \inf_{\gamma: q_0 \rightarrow q_1} L_{\text{resp}}[\gamma],$$

with value  $+\infty$  if no admissible path connects the endpoints.

**Theorem 12.6** (Endpoint process displacement is bounded by response length). *For every absolutely continuous response path,*

$$d_{\text{proc}}(\bar{\Phi}(q_0), \bar{\Phi}(q_1)) \leq L_{\text{resp}}[\gamma],$$

and therefore

$$d_{\text{proc}}(\bar{\Phi}(q_0), \bar{\Phi}(q_1)) \leq d_{\text{resp}}(q_0, q_1).$$

Here  $d_{\text{proc}}(\mathcal{N}_0, \mathcal{N}_1) = \|\mathcal{N}_1 - \mathcal{N}_0\|_{\text{proc}}$  in an affine process chart.

*Proof.* The image path  $\bar{\Phi} \circ \gamma$  is absolutely continuous and has speed  $\|D\bar{\Phi}_{\gamma(t)} \dot{\gamma}(t)\|_{\text{proc}} = F_{\text{resp}}(\gamma(t), \dot{\gamma}(t))$ . The normed-space triangle inequality bounds endpoint displacement by the integrated speed. Taking the path infimum gives the second inequality.  $\square$

**Theorem 12.7** (Finite linear chart and Paper 2 recovery). *Suppose*

$$\Phi(\mathcal{F}, r) = \mathcal{F} + T_{\Pi} r$$

*on a finite-dimensional affine process chart. Put*

$$\bar{R}_{\Pi} := R_{\Pi} / \ker T_{\Pi}, \quad \bar{T}_{\Pi} : \bar{R}_{\Pi} \rightarrow \text{im } T_{\Pi}.$$

*Then the quotient response geometry is flat and translation invariant:*

$$F_{\text{resp}}([r], [v]) = \|\bar{T}_{\Pi}[v]\|_{\text{proc}},$$

*and straight quotient segments minimize length, so*

$$d_{\text{resp}}([r_0], [r_1]) = \left\| \bar{T}_{\Pi}(r_1 - r_0) \right\|_{\text{proc}}.$$

*If  $\|\cdot\|_{\bar{\epsilon}}$  is a calibrated quotient ledger norm, define*

$$L_{\Pi}^{\text{opt}} := \max_{\|x\|_{\bar{\epsilon}}=1} \|\bar{T}_{\Pi}x\|_{\text{proc}}, \quad \mu_{\Pi} := \min_{\|x\|_{\bar{\epsilon}}=1} \|\bar{T}_{\Pi}x\|_{\text{proc}}.$$

*For a nontrivial quotient,*

$$0 < \mu_{\Pi} \leq L_{\Pi}^{\text{opt}} < \infty,$$

*and*

$$\frac{1}{L_{\Pi}^{\text{opt}}} d_{\text{resp}}([r_0], [r_1]) \leq \|r_1 - r_0\|_{\bar{\epsilon}} \leq \frac{1}{\mu_{\Pi}} d_{\text{resp}}([r_0], [r_1]).$$

*Proof.* The quotient removes exactly the kernel of  $T_{\Pi}$ , so  $\bar{T}_{\Pi}$  is injective and the displayed tangent gauge follows from Theorem 12.4. The image of a straight quotient segment is a straight process segment, whose normed-space length equals its endpoint displacement. Theorem 12.6 proves minimality. Compactness of the quotient unit sphere gives the extremal gains, and injectivity makes the minimum positive. Homogeneity gives the final comparison. This is the intrinsic-geometric form of the finite linear response theorem in Paper 2.  $\square$

**Definition 12.8** (Extrinsic free-process monotone). *For a closed free-process set  $\mathfrak{F}_{\text{sep}} \subset \mathcal{P}$ , define*

$$M_{\text{sep}}(\mathcal{N}) := \inf_{\mathcal{F} \in \mathfrak{F}_{\text{sep}}} d_{\text{proc}}(\mathcal{N}, \mathcal{F}).$$

**Proposition 12.9** (Extrinsic monotone versus intrinsic response distance). *If  $\bar{\Phi}(q_0) \in \mathfrak{F}_{\text{sep}}$ , then*

$$M_{\text{sep}}(\bar{\Phi}(q_1)) \leq d_{\text{resp}}(q_0, q_1).$$

*A reverse bound is not automatic. It requires, for example, that a nearest free process lie in the response chart together with a uniform lower response gain and sufficient quasiconvexity of the response image.*

*Proof.* The free-set infimum is no larger than the distance to the particular free point  $\bar{\Phi}(q_0)$ , and Theorem 12.6 bounds that process distance by  $d_{\text{resp}}$ . If the nearest free process lies outside the response image, or if the image has long detours relative to ambient process distance, no converse follows.  $\square$

### 13. Physical-ledger calibration: distance, energy, and the homogeneity fork

**Definition 13.1** (Physical interaction-ledger rate). *A physical path ledger on a response quotient is a nonnegative tangent functional  $j_q(v)$  whose path cost is*

$$J[\gamma] = \int_0^1 j_{\gamma(t)}(\dot{\gamma}(t)) dt.$$

*The induced directed physical path cost is*

$$I_{\Pi}^{\text{path}}(q_0, q_1) := \inf_{\gamma: q_0 \rightarrow q_1} J[\gamma].$$

*The word “physical” refers to a separately calibrated resource valuation, not merely to the represented process norm.*

**Theorem 13.2** (Calibrated ledger realization). *Suppose constants  $0 < a \leq b < \infty$  satisfy*

$$a F_{\text{resp}}(q, v) \leq j_q(v) \leq b F_{\text{resp}}(q, v)$$

*for every admissible tangent direction. Then*

$$\boxed{a d_{\text{resp}}(q_0, q_1) \leq I_{\Pi}^{\text{path}}(q_0, q_1) \leq b d_{\text{resp}}(q_0, q_1)}.$$

*If  $j = F_{\text{resp}}$  exactly, then  $I_{\Pi}^{\text{path}} = d_{\text{resp}}$ . Equality is therefore a calibration theorem, not a definition of interaction cost.*

*Proof.* Integrate the pointwise bounds along every admissible path and take the infimum. Exact calibration makes the two path functionals identical.  $\square$

**Boundary 13.3** (When no path metric exists). Fixed setup burdens, packet thresholds, indivisible tokens, saturation, and history-dependent hysteresis can make physical cost fail local one-homogeneity or refinement additivity. In those branches  $I_{\Pi}^{\text{path}}$  need not be generated by any tangent length. The response distance remains a valid model geometry, but its identification with the physical ledger fails.

**Theorem 13.4** (Local residual energy is squared distance). *Let  $q_0$  be a reference response class and let  $\mathcal{R}$  be a  $C^2$  nonnegative residual penalty with*

$$\mathcal{R}(q_0) = 0, \quad D\mathcal{R}(q_0) = 0.$$

*Assume the Hessian*

$$g_{q_0}^{\text{E}} := D^2\mathcal{R}(q_0)$$

*is positive definite and extends to a smooth local Riemannian metric  $g^{\text{E}}$ . Then*

$$\boxed{\mathcal{R}(q) = \frac{1}{2} d_{g^{\text{E}}}(q_0, q)^2 + o(d_{g^{\text{E}}}(q_0, q)^2)}$$

*as  $q \rightarrow q_0$ .*

*Proof.* Choose normal coordinates for  $g^{\text{E}}$  at  $q_0$ . Taylor expansion gives  $\mathcal{R}(q) = \frac{1}{2} g_{q_0}^{\text{E}}(\Delta q, \Delta q) + o(\|\Delta q\|^2)$ . In the same coordinates, squared geodesic distance is  $g_{q_0}^{\text{E}}(\Delta q, \Delta q) + o(\|\Delta q\|^2)$ . Combining the expansions proves the claim.  $\square$

*Audit Note 13.5* (Interaction cost is not one object). Paper 2 contains both branches. Its quotient coordinate/path-speed bounds are one-homogeneous and naturally define length. Its smooth residual realization gives a two-homogeneous Hessian energy. The correct local dictionary is therefore either

$$I_{\Pi}^{\text{path}} = d_{\text{resp}}$$

under exact path-ledger calibration, or

$$I_{\Pi}^{\text{res}} = \frac{1}{2}d_g^2 + o(d^2)$$

under smooth residual realization. Writing  $I_{\Pi} = d_g$  without specifying the branch loses this distinction.

## 14. Anchoring response geometry to the comparison substrate

Response geometry lives on implementation classes. To obtain substrate geometry one needs a physical map from comparison directions to response directions.

**Definition 14.1** (Universal infinitesimal comparison family). *Fix a regular region  $U \subseteq \mathcal{M}$ . For each  $x \in U$ , let  $\mathcal{A}_x$  denote the full declared family of admitted infinitesimal comparison continuations based at  $x$ . Every  $a \in \mathcal{A}_x$  has a linearized response map*

$$\ell_{a,x} : T_x \mathcal{M} \longrightarrow Y_{a,x},$$

where  $Y_{a,x}$  is the corresponding infinitesimal outcome or process-response space. Define the universal comparison evaluation map

$$E_x : T_x \mathcal{M} \longrightarrow \prod_{a \in \mathcal{A}_x} Y_{a,x}, \quad E_x u := (\ell_{a,x} u)_{a \in \mathcal{A}_x}.$$

No topology on the full product is needed for the kernel statements below. On a finite-rank regular branch one may replace the full family by any finite separating subfamily.

**Definition 14.2** (Contextual null distribution). *Two tangent directions are contextually equivalent exactly when every admitted infinitesimal comparison gives the same response:*

$$u \approx_x v \iff \ell_{a,x}(u - v) = 0 \text{ for every } a \in \mathcal{A}_x.$$

Therefore the contextual null space is not an independent declaration but the kernel of the universal comparison evaluation:

$$N_x^{\text{ctx}} := \ker E_x = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}.$$

The contextual tangent quotient is

$$T_x^{\text{ctx}} \mathcal{M} := T_x \mathcal{M} / N_x^{\text{ctx}}.$$

When  $x \mapsto N_x^{\text{ctx}}$  has locally constant rank, these fibers form a smooth quotient bundle.

**Definition 14.3** (Comparison and physical-response germs). *Let  $\mathcal{C}_x$  be the set of admissible  $C^1$  curve germs  $[\gamma]_x$  through  $x$ . Each admitted comparison  $a \in \mathcal{A}_x$  assigns an outcome germ*

$$\mathcal{L}_{a,x} : \mathcal{C}_x \longrightarrow \mathcal{Y}_{a,x}.$$

The complete comparison-germ evaluation is

$$\mathcal{E}_x([\gamma]_x) := (\mathcal{L}_{a,x}([\gamma]_x))_{a \in \mathcal{A}_x}.$$

Let

$$\mathcal{B}_x^{\text{phys}} : \mathcal{C}_x \longrightarrow \mathcal{P}_x^{\text{red}}$$

be the physical process-response germ valued in the reduced response carrier, obtained after quotienting gauge, auxiliary, and otherwise record-invisible representation variables. Thus  $\mathcal{P}_x^{\text{red}}$  contains only distinctions that are carried by some physical record sector.

**Theorem 14.4** (Closed-world germ-completeness theorem). *Assume the closed-world closure imported from Papers 0 and 1:*

- (i) every finite enforceable distinction belongs to the closed interface ledger and is therefore readable in the complete reduced process response;
- (ii) no stable reduced process direction is absent from every record sector.

Then the complete comparison-germ evaluation and the reduced physical-response germ have the same fibers:

$$\boxed{\mathcal{B}_x^{\text{phys}}([\gamma]_x) = \mathcal{B}_x^{\text{phys}}([\eta]_x) \iff \mathcal{E}_x([\gamma]_x) = \mathcal{E}_x([\eta]_x).}$$

Consequently both maps factor through one and the same contextual quotient

$$\mathcal{Q}_x := \mathcal{C}_x / \sim_{\text{ctx}}, \quad [\gamma]_x \sim_{\text{ctx}} [\eta]_x \iff \mathcal{E}_x([\gamma]_x) = \mathcal{E}_x([\eta]_x),$$

and the induced reduced physical carrier is canonically isomorphic to  $\mathcal{Q}_x$  by the unique class-preserving bijection.

*Proof.* Suppose first that the reduced physical-response germs agree. Any admitted comparison distinguishes a finite cost-bearing partition and therefore belongs to the closed interface ledger. By closed-world closure its outcome is a readout of the complete reduced process response. Equal reduced response germs therefore give equal outcome germs for every  $a$ , so the comparison evaluations agree.

Conversely suppose the comparison evaluations agree but the reduced physical response germs differ. Their difference is then a stable distinction in the reduced process carrier. Because this carrier has already been quotiented by record-invisible redundancy, the distinction must be carried by at least one record sector. The corresponding finite continuation is an admitted comparison in the complete family  $\mathcal{A}_x$ , and it would distinguish the two comparison evaluations, contradiction. Thus the fibers coincide.

Maps with the same fibers induce bijections from their common quotient onto their images. Since  $\mathcal{P}_x^{\text{red}}$  is defined as the complete reduced image, the class-preserving bijection is onto and unique.  $\square$

*Audit Note 14.5* (What closed-world completeness contributes). The theorem does not infer derivative faithfulness directly from ledger closure. It first proves the stronger and correctly typed statement at the level at which Paper 0 defines physical identity: complete families of finite admissible continuations. This avoids the false inference that global separation automatically implies injective differentiation. The latter fails, for example, for  $t \mapsto t^2$  at  $t = 0$ .

**Definition 14.6** (Finite-stage record test). *Let  $W \subseteq P^{\text{red}}$  be a comparison sector, with no topology assumed. A finite-stage record test on  $W$  is a map*

$$r_k : W \longrightarrow S_k,$$

*where  $S_k$  is a finite outcome set and  $r_k$  is produced by a finite admissible continuation and deposited in a physical record sector. The fibres of all such tests form a Boolean algebra of finite operational distinctions. They are not, merely by being executable, declared open sets.*

**Definition 14.7** (Precision-refinement channel and asymptotic quotient). *A precision-refinement channel is a compatible tower*

$$\mathbf{r} = (r_k, \pi_{k+1,k})_{k \geq 0}, \quad r_k : W \rightarrow S_k, \quad r_k = \pi_{k+1,k} \circ r_{k+1},$$

*with every  $S_k$  finite. A real realization consists of encodings  $\iota_k : S_k \rightarrow \mathbb{R}$  for which  $\iota_k r_k$  is locally uniformly Cauchy with certified error  $\epsilon_k \downarrow 0$ . Its limit is*

$$\hat{r} = \lim_{k \rightarrow \infty} \iota_k r_k.$$

*Two complete tower histories are asymptotically record-equivalent when their encoded separation tends to zero. The realization is quotient-faithful when equality of real limits is exactly asymptotic record-equivalence. Thus finite execution is retained at every precision while histories that differ only by vanishing boundary conventions are identified in the reduced channel.*

**Proposition 14.8** (Unreduced tower-faithfulness no-go). *Let  $X_{\mathbf{r}}$  be the inverse limit of the finite discrete stage spaces. If a real realization is continuous and injective on the unreduced tower space  $X_{\mathbf{r}}$ , then its image is compact and zero-dimensional. A finite product of such tower-faithful channel images is again zero-dimensional and cannot contain a nontrivial connected smooth continuation orbit.*

*Proof.* A countable inverse limit of finite discrete spaces is compact, metrizable, and zero-dimensional. A continuous injection from a compact space into Hausdorff  $\mathbb{R}$  is a homeomorphism onto its image, so the image remains zero-dimensional. Finite products preserve zero-dimensionality. Every connected subset is therefore a singleton.  $\square$

**Corollary 14.9** (Positive-dimensional smooth continuum requires more than faithful tower relabelling). *A continuous injective real relabelling of an unreduced profinite tower may exist, but its image remains zero-dimensional. Therefore no finite product of such faithful relabellings can by itself contain a nontrivial connected positive-dimensional smooth continuation orbit. To obtain that orbit, the physical construction must either pass to an asymptotic quotient that changes the relevant topology or add independently certified non-profinite smooth structure. The statement does not exclude real-valued faithful embeddings of Cantor-type tower spaces.*

**Definition 14.10** (Record-limit topology). *Let  $\mathcal{R}_{\text{ch}}(W)$  be the family of ledger-certified real channel limits on  $W$ . The record-limit topology  $\tau_{\text{rec}}$  is the initial topology for the maps*

$$\hat{r} : W \longrightarrow \mathbb{R}, \quad \hat{r} \in \mathcal{R}_{\text{ch}}(W),$$

*where  $\mathbb{R}$  carries its ordinary topology. Equivalently, a subbasis is formed by*

$$\hat{r}^{-1}((a, b)), \quad a, b \in \mathbb{Q}, \quad a < b.$$

*Finite-stage fibres certify finite-resolution distinctions and approximate these neighbourhoods; they are not automatically clopen in  $\tau_{\text{rec}}$ .*

*Boundary 14.11* (Why the finite-stage topology is rejected). If every fibre of every finite-stage test were declared open, the resulting Hausdorff refinement topology would be zero-dimensional and totally disconnected. It could not contain a nontrivial connected smooth continuation orbit. APF therefore uses the finite stages as executable approximants and constructs physical neighbourhoods from the ordered limits of precision channels. This is the exact place where the continuum topology enters; it is not the unrestricted profinite completion of the finite partitions.

**Proposition 14.12** (Finite-stage clopen no-go). *Let  $\tau_{\text{fin}}$  be the topology generated by declaring every finite-stage record fibre open. Then every such fibre is clopen. If the finite stages separate points,  $(W, \tau_{\text{fin}})$  is Hausdorff and zero-dimensional; hence all connected subsets are singletons. No positive-dimensional connected smooth orbit can be embedded continuously in this topology.*

*Proof.* Each stage has finitely many fibres, so the complement of any one fibre is the finite union of the others and is open. Thus the fibre basis is clopen and the space is zero-dimensional. A Hausdorff zero-dimensional space is totally disconnected. The continuous image of a connected manifold is connected, so an injective continuous orbit inclusion would have singleton image unless the orbit were zero-dimensional.  $\square$

**Proposition 14.13** (Closed-world Hausdorff separation). *If the complete real channel family separates reduced physical states, then  $(P^{\text{red}}, \tau_{\text{rec}})$  is Hausdorff.*

*Proof.* Let  $p \neq q$ . Closed-world reduction supplies a real precision channel  $\hat{r}$  with  $\hat{r}(p) \neq \hat{r}(q)$ . Choose disjoint rational open intervals containing the two values. Their inverse images are disjoint  $\tau_{\text{rec}}$ -neighbourhoods of  $p$  and  $q$ .  $\square$

**Assumption 14.14** (Countable channel-specification grammar). *The primitive instruction tokens, finitely encoded exact parameters, stopping rules, bonding-map descriptions, encodings, and error certificates available on the declared interface form a countable specification language. Arbitrary real constants are not admitted as single primitive tokens unless an explicit countable code is supplied.*

**Definition 14.15** (Finitely specified physical channel). *A physical precision channel is finitely specified when one finite word in the grammar of Assumption 14.14 specifies its stage tests, bonding maps, requested-precision stopping rule, encodings, and certified error bound.*

**Theorem 14.16** (Countable specified channel family). *Under Assumption 14.14, the finitely specified physical precision channels on a fixed closed interface form a countable family.*

*Proof.* The set of finite words over a countable alphabet is countable, and each specified channel is determined by one such word.  $\square$

**Gate 14.17** (Countable topology generation). *The finitely specified quotient-faithful real channel limits contain a countable subfamily whose initial topology equals the record-limit topology defined by the complete ledger-certified channel family.*

**Corollary 14.18** (Conditional second countability). *Under Assumption 14.14 and Gate 14.17, the closed-interface record-limit topology is second countable.*

*Proof.* Inverse images of rational open intervals under the countable topology-generating subfamily form a countable subbasis; finite intersections form a countable base.  $\square$

**Lemma 14.19** (Finite stages approximate record-limit neighbourhoods). *Let  $\hat{r}$  be a channel limit with certified local error  $|\iota_k r_k - \hat{r}| \leq \epsilon_k$ . For every  $p$  and every open interval  $I \ni \hat{r}(p)$ , there are a smaller interval  $J$  with  $\hat{r}(p) \in J \subseteq I$  and a stage  $k$  such that the finite-stage cell containing  $p$  is*

mapped by  $\iota_k$  into the  $\epsilon_k$ -thickening of  $J$ , which lies in  $I$ . Hence every record-limit neighbourhood is operationally resolvable to finite certified precision.

*Proof.* Choose  $J$  with positive distance  $\delta$  from its closure to  $\mathbb{R} \setminus I$ . Take  $k$  with  $\epsilon_k < \delta/2$  and with  $\iota_k r_k(p)$  within  $\epsilon_k$  of  $\hat{r}(p)$ . Compatibility and the certified error bound place all states in the corresponding stage cell whose encoded value lies in the chosen thickening inside  $I$ . No claim that the entire stage cell is open is required.  $\square$

**Definition 14.20** (Executable channel algebra). *For  $U \in \tau_{\text{rec}}$ , let  $\mathcal{A}_{\text{ch}}(U)$  be the unital real algebra generated by restrictions of real precision-channel limits and by ledger-certified locally uniform limits of executable scalar postprocessings. Every generator is  $\tau_{\text{rec}}$ -continuous by construction or by uniform convergence. Let  $\mathcal{A}_0(U)$  denote its closure under such locally uniform limits.*

*Audit Note 14.21* (Topology ledger). The finite stages provide the finite executable evidence. The real channel limits provide the ordered topology. Closed-world completeness provides point separation. A countable specification grammar provides a countable specified channel family; second countability follows only when that family is topology-generating. None of these statements identifies the raw inverse limit of finite partitions with a continuum, and none assumes a manifold.

**Definition 14.22** (Executable postprocessing and smooth saturation). *Let  $f = (f_1, \dots, f_m)$  be a jointly realizable finite channel map at the requested precision. A scalar postprocessing is executable when it is given by a finite protocol at each requested precision. It is observationally conservative when it factors through  $f$ , and therefore cannot distinguish two states that  $f$  does not already distinguish.*

*The smooth saturation of a family of real readouts is its closure under all maps  $\psi \in C^\infty(\mathbb{R}^m)$ . Smooth saturation is a mathematical closure of the represented observable algebra; it does not assert that every smooth map is itself a primitive finite program.*

**Theorem 14.23** (Lossless postprocessing factor theorem). *Every executable deterministic postprocessing of a closed joint record is observationally conservative. Conversely, adjoining any scalar function already constant on the fibres of the closed joint record changes neither those fibres nor the closed-world contextual quotient. No executability claim is implied in this converse.*

*Proof.* If  $h = \varphi \circ f$ , then  $f(p) = f(q)$  implies  $h(p) = h(q)$ , so  $h$  creates no new distinction class. Appending an executable protocol to the record continuation changes implementation cost but not the partition of physical states induced by the complete record. Conversely, any scalar function constant on every fibre of  $f$  factors set-theoretically through the image of  $f$ ; adjoining it leaves those fibres unchanged. Hence the contextual quotient is invariant under such lossless relabelling.  $\square$

**Definition 14.24** (Refinement-complete APF differential sheaf). *For  $U \in \tau_{\text{rec}}$ , let  $\mathcal{R}(U)$  contain the refinement-stable real readouts obtained from executable records and their ledger-certified local limits. Define  $\mathcal{C}_{\text{APF}}(U)$  to be the localization and smooth saturation of  $\mathcal{R}(U)$ : it is the smallest family of real functions on  $U$  that*

- (i) contains  $\mathcal{R}(U)$ ;
- (ii) is closed under restriction and local gluing;
- (iii) satisfies

$$f_1, \dots, f_m \in \mathcal{C}_{\text{APF}}(U), \quad \psi \in C^\infty(\mathbb{R}^m) \implies \psi(f_1, \dots, f_m) \in \mathcal{C}_{\text{APF}}(U).$$



**Theorem 14.25** (Differential-space closure theorem). *The triple*

$$(P^{\text{red}}, \tau_{\text{rec}}, \mathcal{C}_{\text{APF}})$$

*is a Sikorski differential space. Its topology is the initial topology of the physical readouts, and its differential structure is closed under smooth superposition and localization.*

*Proof.* The record-limit topology is, by Definition 14.10, the initial topology of the generating physical channel limits. Smooth superpositions of continuous generators remain continuous, and localization adds only functions that locally agree with such continuous functions. Definition 14.24 is therefore closed under smooth superposition and localization and induces the same topology. These are the Sikorski axioms. Proposition 14.13 and Proposition 14.18 supply Hausdorffness and, on countably generated sectors, second countability; neither property is needed to define the differential structure itself. The lossless-postprocessing factor theorem shows that the saturation changes no closed-world record fibre.  $\square$

*Audit Note 14.26* (What is derived and what is saturated). The executable channel algebra, its record-limit topology, and invariance of record fibres under deterministic postprocessing are derived from APF closure. Closure under the full  $C^\infty$  functional calculus is the canonical smooth saturation of that algebra. It is not disguised computability and it does not assert that finite records alone force a manifold. Its role is to define the minimal differential structure compatible with the already represented readouts; manifold structure appears only on the regular continuation orbits below.

**Definition 14.27** (Derivation tangent and algebraic cotangent). *Let  $\mathcal{C}_p$  be the stalk of  $\mathcal{C}_{\text{APF}}$  at  $p$ , and let  $\mathfrak{m}_p = \{[f]_p : f(p) = 0\}$  be its maximal ideal. The primary tangent is*

$$T_p^{\text{der}} P := \text{Der}_p(\mathcal{C}_p, \mathbb{R}),$$

*the real derivations satisfying Leibniz' rule. The algebraic first-order cotangent candidate is*

$$T_p^{*,\text{alg}} P := \mathfrak{m}_p / \mathfrak{m}_p^2.$$

*Every derivation defines a linear functional on  $\mathfrak{m}_p / \mathfrak{m}_p^2$ ; Proposition 14.28 shows that this pointwise pairing is canonical and perfect. Singular behaviour appears through rank variation and failure of finite local presentation.*

**Proposition 14.28** (Canonical tangent–cotangent duality). *Evaluation on first-order classes gives a canonical vector-space isomorphism*

$$T_p^{\text{der}} P \cong (\mathfrak{m}_p / \mathfrak{m}_p^2)^*, \quad X \mapsto ([f] \mapsto Xf).$$

*Singularity is therefore detected by failure of finite local presentation, rank jumps, or variation of this common dimension—not by a spurious gap between the two objects.*

*Proof.* Leibniz' rule makes every derivation vanish on  $\mathfrak{m}_p^2$ , so the map is well defined and injective. Conversely, for  $\lambda \in (\mathfrak{m}_p / \mathfrak{m}_p^2)^*$  define

$$X_\lambda(f) := \lambda([f - f(p)]).$$

Modulo  $\mathfrak{m}_p^2$  one has  $fg - f(p)g(p) = f(p)(g - g(p)) + g(p)(f - f(p))$ , hence  $X_\lambda$  satisfies Leibniz' rule at  $p$ . The two constructions are inverse.  $\square$

**Definition 14.29** (Finite differential presentation). *A record-open neighbourhood  $U$  has a finite differential presentation when there exist real channel limits  $f_1, \dots, f_N \in \mathcal{C}_{\text{APF}}(U)$  such that*

(i) the evaluation map

$$F_U = (f_1, \dots, f_N) : U \rightarrow \mathbb{R}^N$$

is injective and is a homeomorphism onto its image;

(ii) every  $g \in \mathcal{C}_{\text{APF}}(U)$  is locally of the form  $g = \psi \circ F_U$  for some  $\psi \in C^\infty(\mathbb{R}^N)$ .

The locus of points admitting such a neighbourhood is denoted  $P^{\text{fdiff}}$ .

**Definition 14.30** (Ledger-independent standing channels). A jointly realizable family of precision channels  $\mathbf{r}^1, \dots, \mathbf{r}^m$  is ledger-independent on  $U$  when no complete tower  $\mathbf{r}^i$  is determined by the joint complete tower of the others. Equivalently, for every  $i$  there exist reduced states  $p_i, q_i \in U$  that agree in all channels  $j \neq i$  but differ in channel  $i$ . A ledger-independent channel therefore carries a distinction not recoverable from the remaining standing commitments.

**Theorem 14.31** (Additive channel-capacity bound). Assume the APF positive standing-cost floor  $\varepsilon_* > 0$ , additive ledger accounting for jointly realized ledger-independent commitments, and finite local capacity  $C(U) < \infty$ . Then every ledger-independent jointly realizable family of precision channels on  $U$  has cardinality

$$m \leq \left\lfloor \frac{C(U)}{\varepsilon_*} \right\rfloor.$$

The bound concerns the number of independent standing channels, not the number of finite-stage outcomes, the number of states, or the cardinality of any refinement limit.

*Proof.* Ledger independence assigns to each channel a distinction unavailable from the others. By the positive floor, maintaining each such standing distinction costs at least  $\varepsilon_*$ . Additivity for simultaneously maintained independent commitments therefore gives total standing charge at least  $m\varepsilon_*$ . Admissibility requires  $m\varepsilon_* \leq C(U)$ .  $\square$

**Definition 14.32** (Complete channel basis). A complete channel basis on  $U$  is a jointly realizable channel family that separates reduced states and is inclusion-minimal with that property. Equivalently, deleting any one channel destroys separation.

**Lemma 14.33** (Minimal completeness implies ledger independence). Every complete channel basis is ledger-independent.

*Proof.* If channel  $i$  were determined by all the others, then equality in the remaining channels would force equality in channel  $i$ . Deleting channel  $i$  would therefore preserve the joint fibres and hence preserve separation, contradicting inclusion-minimality.  $\square$

**Theorem 14.34** (Finite-channel completeness). Let  $U$  admit a complete channel basis and satisfy the additive channel-capacity theorem. Then every complete channel basis is finite:

$$\mathbf{F}_U = (\mathbf{r}^1, \dots, \mathbf{r}^N), \quad N \leq \left\lfloor \frac{C(U)}{\varepsilon_*} \right\rfloor.$$

Its joint real limit map

$$\hat{F}_U = (\hat{r}^1, \dots, \hat{r}^N) : U \rightarrow \mathbb{R}^N$$

is injective whenever the chosen channels possess faithful real realizations.

*Proof.* By Lemma 14.33, a complete channel basis is ledger-independent. The additive capacity bound therefore makes it finite. Completeness says that equality of every complete channel tower implies equality in the reduced carrier. Faithful real realization preserves each tower's complete

fibres; hence equality of all real limits implies equality of all channel towers and therefore equality of states. Thus  $\widehat{F}_U$  is injective.  $\square$

**Gate 14.35** (Finite closed-interface generation). *Finite capacity bounds every jointly realizable ledger-independent channel family, but it does not by itself prove that one finite family separates the entire reduced sector. The missing statement is the finite-generation theorem*

*closed-world completeness + finite capacity  $\implies$  finite complete physical channel basis.*

*It requires either a channel-dependence exchange theorem, a compact finite-subcover theorem for physical distinctions, or an equivalent APF generation result. Without one of these, a countable separating family can require infinitely many channels even though no large independent family is jointly realizable.*

**Gate 14.36** (Faithful operational-to-precision-channel realization). *A closed interface  $U$  closes the operational-to-channel bridge when there is a map*

$$\mathfrak{R}_U : \mathcal{A}_U \longrightarrow \text{Ch}_U$$

*from operational classes to physical precision-channel towers satisfying:*

(i) joint preservation:  $S \in \mathfrak{J}_U$  implies  $\mathfrak{R}_U(S)$  is jointly realizable;

(ii) closure preservation and reflection:

$$[a]_U \in \text{cl}_U(S) \iff \mathfrak{R}_U([a]_U) \text{ is deterministically recoverable from } \mathfrak{R}_U(S);$$

(iii) state completeness: *the full image family separates reduced physical states; and*

(iv) faithful real realization: *each complete channel tower has a real limit with the same complete fibres.*

**Theorem 14.37** (Conditional finite precision-channel basis). *If Theorem 2.14 and Gate 14.36 hold, then the image of a finite complete operational basis is a finite jointly realizable separating precision-channel family. Deleting channel-redundant members yields a complete channel basis with cardinality at most  $\lfloor C_U/\varepsilon_* \rfloor$ .*

*Proof.* Joint preservation gives simultaneous realizability. Closure preservation and state completeness give separation. Finiteness and the capacity bound are inherited from the operational basis. Redundant deletion preserves joint realizability because the joint domain is downward closed.  $\square$

**Audit Note 14.38** (Bridge disposition). The theorem above is a proved implication. The physical realization map  $\mathfrak{R}_U$  remains a named gate. Therefore the operational basis theorem does not silently close finite physical channel generation.

**Proposition 14.39** (Finite-channel sufficiency versus finite differential sufficiency). *Theorem 14.34 gives a finite separating channel map  $\widehat{F}_U$ . It gives a finite differential presentation exactly when*

(i)  $\widehat{F}_U$  is a homeomorphism from  $U$  onto its image in the record topology; and

(ii) every  $g \in \mathcal{C}_{\text{APF}}(U)$  factors locally as  $g = \psi \circ \widehat{F}_U$  with  $\psi \in C^\infty(\mathbb{R}^N)$ .

*Proof.* Separation gives injectivity. The topology and smooth-factorization clauses are precisely the two additional conditions in Definition 14.29. No finite-image inference is used.  $\square$

**Theorem 14.40** (Subcartesian realization criterion). *Every point of  $P^{\text{fdiff}}$  has a neighbourhood isomorphic, as a differential space, to a differential subspace of a subset of  $\mathbb{R}^N$ . Hence  $P^{\text{fdiff}}$  is locally subcartesian. Conversely, any locally subcartesian presentation by physical readouts is a finite differential presentation in the sense above.*

*Proof.* Condition (i) identifies  $U$  homeomorphically with  $F_U(U)$ . Condition (ii) says that the differential structure on  $U$  is exactly the pullback of the restriction of  $C^\infty(\mathbb{R}^N)$  to that image. This is the defining local model of a subcartesian differential space. The converse is immediate from a finite ambient-coordinate presentation.  $\square$

**Gate 14.41** (Finite smooth generation). *Even after a finite state-separating channel family is obtained, finite smooth generation is a distinct theorem. It requires that the continuation-regular readout sheaf be locally the smooth functional calculus of that finite map:*

$$\mathcal{C}_{\text{APF}}|_U = \text{Loc } C^\infty(f_1, \dots, f_N).$$

*State separation alone does not imply this equality. Raw threshold records may be discontinuous, and continuous limit readouts need not be smooth functions of a separating map. The subcartesian orbit theorem is invoked only after this finite smooth-generation statement closes.*

*Audit Note 14.42* (Open-item ledger before regular response geometry). No topological or smooth conclusion is promoted past its antecedent. The remaining substantive certificates are:

- (i) faithful realization of the finite operational basis as a jointly realizable, state-separating precision-channel family;
- (ii) a countable specification grammar and countable topology-generating subfamily whenever second countability is claimed;
- (iii) finite smooth generation of the continuation-regular readout sheaf;
- (iv) a locally flowable continuation family complete in the orbit-theorem sense; and
- (v) smooth two-way response-factor completeness before the derivative-kernel identity is used.

A physical comparison-substrate anchor is a further downstream gate after intrinsic response-orbit geometry has been obtained.

*Remark 14.43* (Imported singular differential geometry). For subcartesian differential spaces, derivations satisfy the smooth chain rule. The orbits of a family complete in the orbit-theorem sense carry unique smooth manifold structures for which the orbit inclusions are smooth immersions. The paper imports these established results rather than reproving them. See Sikorski’s differential-space formalism and the subcartesian orbit results of Śniatycki and collaborators.<sup>1</sup>

**Definition 14.44** (Continuation distribution and geometric orbits). *Let  $\mathcal{X}_{\text{adm}}$  be a family of vector fields on  $P^{\text{fdiff}}$ , where a vector field is a derivation that generates a local one-parameter group of local diffeomorphisms. The family is complete in the orbit-theorem sense when for every  $X, Y \in \mathcal{X}_{\text{adm}}$  and every time  $t$  for which the local flow  $\varphi_t^X$  is defined, the pushforward  $(\varphi_t^X)_*Y$  agrees locally with*

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<sup>1</sup>J. Śniatycki, *Orbits of families of vector fields on subcartesian spaces*, Ann. Inst. Fourier 53 (2003), 2257–2296, DOI: 10.5802/aif.2006; J. Śniatycki, *Differential Geometry of Singular Spaces and Reduction of Symmetry*, Cambridge University Press (2013), DOI: 10.1017/CBO9781139136990; R. Cushman and J. Śniatycki, *On subcartesian spaces* *Leibniz’ rule implies the chain rule*, Can. Math. Bull. 63 (2020), DOI: 10.4153/S0008439519000513.

a member of  $\mathcal{X}_{\text{adm}}$ . Equivalently, the family is closed locally under the flow transports used to form its pseudogroup. Define

$$\Delta_p := \text{span}\{X(p) : X \in \mathcal{X}_{\text{adm}}\} \subseteq T_p^{\text{der}} P.$$

The orbit through  $p$  under finite compositions of these local flows is denoted  $\mathcal{O}_p$ .

**Theorem 14.45** (Smooth geometric-orbit theorem). *Every orbit  $\mathcal{O}_p$  of an admissible continuation family complete in the orbit-theorem sense on  $P^{\text{fdiff}}$  carries a unique smooth manifold structure for which its inclusion into the differential space is a smooth immersion. Its tangent space is the continuation distribution:*

$$T_q \mathcal{O}_p = \Delta_q \quad (q \in \mathcal{O}_p).$$

Thus smooth manifolds arise as the dynamically connected regular pieces of the APF differential space, not as prior substrate assumptions.

*Proof.* By Theorem 14.40,  $P^{\text{fdiff}}$  is locally a subcartesian differential space. The source-level completeness clause in Definition 14.44 is exactly the closure hypothesis required by the orbit theorem for complete families of vector fields on subcartesian spaces; hence it applies to  $\mathcal{X}_{\text{adm}}$  and makes each orbit a smooth manifold with the stated tangent distribution.  $\square$

**Definition 14.46** (Maximal smooth geometric branch and singular locus). *A smooth geometric branch is a maximal connected continuation orbit on which  $\dim \Delta$  is locally constant and positive. The union of all such orbits is  $P_{\text{reg}}^{\text{geom}}$ . Its complement*

$$P^{\text{sing}} := P^{\text{red}} \setminus P_{\text{reg}}^{\text{geom}}$$

*is the intrinsic singular locus. It includes zero-rank, rank-jump, saturation, branching, and non-flowable points.*

**Theorem 14.47** (Existence of the regular geometric regime). *Let  $p \in P^{\text{fdiff}}$ . If the admissible continuation family is complete in the orbit-theorem sense near  $p$ , continuation-active at  $p$ , and has locally constant positive orbit rank, then there is a finite-dimensional smooth geometric branch through  $p$ .*

*Proof.* The subcartesian realization criterion supplies the required local singular smooth model. Continuation activity gives a nonzero flow-generating vector field, and Theorem 14.45 gives an immersed smooth orbit through  $p$ . Local constancy of  $\dim \Delta$  fixes its dimension, while maximal connected extension produces the branch.  $\square$

**Theorem 14.48** (Consolidated APF regular-geometry theorem). *Let  $U \subseteq P^{\text{red}}$  be a closed-world comparison domain satisfying: a quotient-faithful state-separating real channel family; the countable grammar and topology-generation conditions of Assumption 14.14 and Gate 14.17; a complete finite channel basis; finite smooth generation; and a locally flowable continuation family complete in the orbit-theorem sense. Then:*

- (i) *the channel limits generate a Hausdorff second-countable record topology;*
- (ii) *localization and smooth saturation produce a Sikorski differential space;*
- (iii) *on  $U \cap P^{\text{fdiff}}$  the differential space is locally subcartesian;*
- (iv) *every orbit of the complete admissible-continuation family is an immersed smooth manifold; and*

(v) on each constant-rank orbit  $S$  satisfying Gate 14.53,

$$\ker D\beta_x = N_x^{\text{ctx}} = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}, \quad x \in S,$$

and the induced map

$$T_x S / N_x^{\text{ctx}} \hookrightarrow T_{\beta(x)} \mathcal{O}_{\beta(x)}$$

is injective.

Thus ordinary smooth response geometry is the regular orbit locus of the APF record differential space, not an upstream manifold postulate.

*Proof.* Item (i) is Propositions 14.13 and 14.18. Items (ii)–(iv) are Theorem 14.25, Theorem 14.40, and Theorem 14.45. Item (v) is Theorem 14.54 and Corollary 14.55.  $\square$

*Audit Note 14.49* (Resolution of the continuum-emergence objection). The raw refinement completion remains profinite and may be totally disconnected. Smooth continuum is not obtained by declaring its points close. It appears on the finite-differential-presentation locus as an orbit of physically admissible local continuation derivations. Thus two independent conditions are visible rather than hidden: smooth local factorization of the record algebra, and continuation-active local flows. Failure of either leaves a singular, higher-order, or zero-dimensional record sector rather than a smuggled manifold.

*Audit Note 14.50* (Fortification ledger: what remains conditional). The differential-space architecture is noncircular, but the following obligations remain deliberately named:

- (i) faithful real channel realization and state separation;
- (ii) a countable topology-generating channel subfamily when second countability is claimed;
- (iii) a complete channel basis, obtained here only after the operational-to-channel realization gate closes;
- (iv) finite smooth generation and the homeomorphism/local-factorization clauses needed for a subcartesian presentation;
- (v) a local flow family complete in the orbit-theorem sense; and
- (vi) smooth response-factor completeness before set-level fibre equivalence is promoted to a derivative-kernel identity.

These are falsifiable regime certificates. None is replaced by the statement that finite partitions alone produce a continuum.

**Definition 14.51** (Contextual comparison differentials). *On a smooth geometric branch  $S$ , every admitted regularized comparison  $L_a \in \mathcal{C}_{\text{APF}}$  has differential*

$$\ell_{a,x} := d(L_a|_S)_x : T_x S \rightarrow Y_{a,x}.$$

Define

$$N_x^{\text{ctx}} := \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x}.$$

Let  $\beta : S \rightarrow P^{\text{red}}$  be the complete reduced physical-response map and  $B_x^{\text{phys}} := D\beta_x$  on the orbit manifold.

*Audit Note 14.52* (Same fibres do not imply the same derivative kernel). On  $\mathbb{R}$ , the maps  $L(t) = t$  and  $\beta(t) = t^3$  have identical fibres, but  $dL_0 \neq 0$  while  $D\beta_0 = 0$ . Their set-theoretic factor is not smooth at the origin. Closed-world fibre completeness is therefore insufficient for the first-order kernel identity.

**Gate 14.53** (Smooth response-factor completeness). *On a neighbourhood  $V \subseteq S$  of  $x$ , there is a finite APF readout generator*

$$F = (f_1, \dots, f_N) : V \rightarrow \mathbb{R}^N$$

*whose components belong to the admitted readout algebra and generate the local Sikorski structure, together with smooth maps on the relevant images such that*

$$F = \Psi \circ \beta, \quad \beta = \Theta \circ F.$$

*Equivalently, the complete reduced response and a finite local APF generator are smoothly equivalent at first order on the declared regular branch.*

**Theorem 14.54** (Regular-orbit closed-world kernel theorem — corrected). *On a smooth geometric branch satisfying closed-world germ completeness and Gate 14.53,*

$$\ker D\beta_x = \ker DF_x = \bigcap_{a \in \mathcal{A}_x} \ker \ell_{a,x} = N_x^{\text{ctx}}.$$

*Proof.* Differentiating  $F = \Psi \circ \beta$  and  $\beta = \Theta \circ F$  gives  $\ker D\beta_x = \ker DF_x$ . Every admitted readout is locally a smooth function of  $F$ , so  $\ker DF_x$  is contained in every admitted readout kernel. Conversely the components  $f_i$  are themselves admitted readouts, so a vector annihilated by every admitted readout differential lies in  $\ker DF_x$ .  $\square$

**Corollary 14.55** (Faithful quotient response). *The induced map*

$$\tilde{B}_x : T_x S / N_x^{\text{ctx}} \longrightarrow T_{\beta(x)} \mathcal{O}_{\beta(x)}, \quad [u] \longmapsto B_x^{\text{phys}} u,$$

*is injective.*

**Definition 14.56** (Universal operational response carrier). *For  $x \in S$ , define*

$$T_x^{\text{op}} S := T_x S / N_x^{\text{ctx}}.$$

*This quotient is universal among linear first-order response realizations that annihilate every contextually null direction.*

**Theorem 14.57** (Exact-realization criterion for a concrete response model). *Let  $B_x^{\text{raw}} : T_x S \rightarrow W_x$  be a concrete finite response model. The following are equivalent:*

- (i)  $\ker B_x^{\text{raw}} = N_x^{\text{ctx}}$ ;
- (ii)  $B_x^{\text{raw}}$  factors through an injective map  $T_x^{\text{op}} S \hookrightarrow W_x$ ;
- (iii) *the model is sound on contextual null directions and jointly separating on the quotient.*

*Proof.* This is the first isomorphism theorem applied to the quotient map  $T_x S \rightarrow T_x S / N_x^{\text{ctx}}$ .  $\square$

**Theorem 14.58** (Canonical response-orbit quotient gauge). *Let  $S$  be a smooth geometric branch and suppose the process carrier has norm  $\|\cdot\|_{\text{proc}}$ . Then*

$$c_x^{\text{op}}([u]) := \|\tilde{B}_x[u]\|_{\text{proc}}$$

defines a positive homogeneous gauge on  $T_x^{\text{op}}S = T_xS/N_x^{\text{ctx}}$ . If the response norm is smooth and strongly convex away from zero, these gauges form a Finsler structure on each locally constant-rank orbit stratum. This gauge is intrinsic to the response orbit; it is not yet a map from physical site displacements.

*Proof.* Well-definedness follows from quotienting by the kernel; positivity follows from injectivity of  $\tilde{B}_x$ . Homogeneity and convexity pull back from the process norm. Smooth bundle dependence follows on constant-rank orbit strata.  $\square$

**Corollary 14.59** (Direct pullback case). *When  $\beta$  itself is the complete response map, the operational gauge is the direct pullback of the process norm by  $D\beta$  after quotienting its kernel.*

**Gate 14.60** (Physical comparison-substrate anchor). *There exist a physical regular comparison domain  $M_{\text{reg}}$ , a smooth site map  $s : M_{\text{reg}} \rightarrow S$ , and a locally constant-rank bundle map*

$$A_x : T_x M_{\text{reg}} \longrightarrow T_{s(x)}^{\text{op}} S$$

*such that, writing  $q_y : T_y S \rightarrow T_y^{\text{op}} S = T_y S / N_y^{\text{ctx}}$  for the quotient map:*

- (i)  *$\ker A_x$  is exactly the physical comparison-null subspace;*
- (ii) *the map has declared process/ledger provenance and is invariant under response-coordinate gauge;*
- (iii) *the kinematic anchor is the quotient differential of the site map,*

$$A_x = q_{s(x)} \circ Ds_x;$$

(iv) *the direct comparison germ is calibrated by  $c_x^{\text{dir}}(u) = c_{s(x)}^{\text{op}}(A_x u)$ ; and*

(v) *the calibration is stable on the declared regular branch.*

**Theorem 14.61** (Conditional physical substrate pullback). *Write  $L_{\text{orb}}[\delta] := \int c_{\delta(t)}^{\text{op}}([\dot{\delta}(t)]) dt$  for the response-orbit length (Theorem 14.58) of a piecewise  $C^1$  curve  $\delta$  in  $S$ . If Gate 14.60 holds, define  $c_x^{\text{sub}}(u) := c_{s(x)}^{\text{op}}(A_x u)$ . Then every piecewise  $C^1$  curve  $\gamma$  in  $M_{\text{reg}}$  satisfies*

$$L_{\text{sub}}[\gamma] := \int c_{\gamma(t)}^{\text{sub}}(\dot{\gamma}(t)) dt = L_{\text{orb}}[s \circ \gamma].$$

*Thus response-orbit geometry becomes physical comparison-substrate geometry only after the anchor gate closes.*

*Proof.* By Gate 14.60(iii),

$$A_{\gamma(t)} \dot{\gamma}(t) = q_{s(\gamma(t))} (Ds_{\gamma(t)} \dot{\gamma}(t)) = [(s \circ \gamma)'(t)].$$

Substitution into the definition of  $c^{\text{sub}}$  gives equality of the two integrands, hence equality of the lengths. Clauses (i), (ii), (iv), and (v) supply the physical null-space, provenance, calibration, and regularity conditions.  $\square$

**Proposition 14.62** (Positive cost metric from quadraticity). *If the process norm satisfies the parallelogram identity on the image of every  $\tilde{B}_x$ , then polarization defines a positive-definite inner product on  $T_x^{\text{op}}S$ . These inner products form the positive cost metric  $g^{\text{cost}}$  on each smooth constant-rank branch.*



*Proof.* The Jordan–von Neumann theorem turns the quotient norm into an inner product. Injectivity makes it positive definite; smooth dependence follows from the smooth response representation.  $\square$

**Definition 14.63** (Comparison atlas). *The atlas used below is the maximal smooth atlas of the continuation orbits in  $P_{\text{reg}}^{\text{geom}}$ . It is derived from the APF differential space and its admissible-flow orbits, not assumed on the full reduced carrier.*

## 14.1 What finite-ledger assumptions become geometrically

The finite-ledger assumptions now divide cleanly. Closed-world completeness provides point separation and complete readout fibres. Finite instruction provides second countability. Lossless postprocessing preserves those fibres, and smooth saturation supplies the canonical Sikorski functional calculus. Finite capacity provides finite-channel sufficiency; smooth local factorization is the separate, explicitly named finite-differential-sufficiency gate. Admissible continuation then supplies a family complete in the orbit-theorem sense of flow-generating derivations, whose orbits are immersed smooth manifolds. Constant rank supplies the quotient bundle used by the response geometry.

## 14.2 Failure witnesses

The differential-space formulation makes failure informative:

- a Stone-like zero-dimensional sector has records but no continuation orbit of positive dimension;
- a rank jump is an intrinsic singular stratum;
- failure of finite-channel sufficiency signals saturation or unbounded local capacity;
- failure of finite differential sufficiency signals nonsmooth local factorization even when the physical record remains complete;
- a jump in the common derivation/cotangent dimension is a singularity witness;
- failure of concrete-model separation means only the numerical model, not the universal physical response, is incomplete.

## 15. Positive cost geometry and Lorentzian comparison geometry

The pullback construction above is positive. Physics also requires a signed causal comparison structure. These are related structures, but they are not the same tensor obtained under different notation.

**Proposition 15.1** (Positive pullbacks cannot produce Lorentzian signature). *Let  $h$  be a positive semidefinite bilinear form on a process tangent space and  $B : T_x\mathcal{M} \rightarrow T_{\mathcal{N}}\mathcal{P}$  linear. Then  $B^*hB$  is positive semidefinite. In particular, no construction of the form  $T^*T$  with a positive ambient inner product can have Lorentzian index one.*

*Proof.* For every  $u$ ,  $(B^*hB)(u, u) = h(Bu, Bu) \geq 0$ . A Lorentzian form has vectors of both signs, so it cannot arise in this way.  $\square$

**Principle 15.2** (Causal cone from finite coordination). *If comparison continuations have finite maximum coordination throughput, tangent directions separate into causally admissible, null-boundary,*

and unreachable classes. A smooth represented regime should encode these classes by a cone field  $C_x \subset T_x\mathcal{M}$ .

**Principle 15.3** (Temporal orientation from irreversible record order). *Papers 1, 3, and 37 supply the distinction between coherent held identity and irreversible record-bearing order. Only after a causal cone field has been independently constructed may Paper 9 represent that record preorder as a time orientation of the cone. Record irreversibility therefore selects between the two components of an already available causal cone; it does not by itself derive Lorentzian signature or the cone.*

**Gate 15.4** (Signed Lorentz–Finsler comparison gate). *A signed comparison representation is available when there is a continuous two-homogeneous function*

$$Q_x^{\text{cmp}} : T_x\mathcal{M} \longrightarrow \mathbb{R}$$

*that is  $C^2$  on the relevant nonzero conic domains, has null set equal to the boundary of the finite-coordination cone, and whose vertical Hessian*

$$g_{x,v}^{\text{cmp}} := \frac{1}{2} D_v^2 Q_x^{\text{cmp}}$$

*has Lorentzian index one. If this Hessian is independent of  $v$ , then*

$$Q_x^{\text{cmp}}(v) = q_x(v, v)$$

*for a Lorentzian bilinear form  $q_x$ .*

**Gate 15.5** (Cost–causal bridge). *The positive operational cost gauge  $c_x^{\text{anc}}$  and the signed comparison function  $Q_x^{\text{cmp}}$  must be related by an explicit physical dictionary. Possibilities include a common normalization on selected timelike or spacelike probes, a bound between implementation cost and  $\sqrt{|Q^{\text{cmp}}|}$ , or a variational relation. No universal identity is assumed, especially on null directions.* **Boundary 15.6** (Why sectorwise polarization is insufficient). *Timelike and spacelike cones are not linear subspaces. Imposing a parallelogram identity separately on positive- and negative-sign sectors does not by itself reconstruct the cross-sector bilinear pairing or a global Lorentzian form. The signed two-homogeneous function or an equivalent cone Lagrangian must be defined on a common tangent domain before its Hessian can certify signature.*

**Gate 15.7** (Dimension gate). *The dimension gate closes when the number of independent local comparison modes is fixed by substrate capacity, representation stability, admissible interaction structure, and the comparison-response anchor. A scalar-register lower bound may constrain one implementation architecture, but it is not by itself a theorem that the substrate tangent bundle has dimension four. This draft continues to treat  $(1 + 3)$  as an input to the weak-field application.*

**Audit Note 15.8** (Two geometric tensors). *In the quadratic branch the paper may carry both*

$$g_x^{\text{cost}} \quad \text{and} \quad q_x^{\text{cmp}}.$$

*The first is positive and measures calibrated response effort. The second is signed and encodes causal comparison. They may share preferred directions or be linked by the recruitment dynamics, but identifying them requires a theorem beyond the pullback construction.*

## 16. The geometric recruitment functional

The geometric substrate is not a passive manifold. It is the substrate that carries comparison continuations. Matter and stress-energy distinctions recruit that substrate; the recruitment load changes the future cost of comparison. The functional below is the geometric specialization of the recruitment cost functional of Papers 24/25.

**Definition 16.1** (Effective geometric recruitment load). *Let  $\chi_{\text{rec}}(x)$  denote the effective accumulated recruitment load carried by the geometric substrate at  $x$ . The notation  $\chi$ , rather than  $\rho$ , is deliberate: the leading exterior point-mass profile is potential-like, scaling as  $1/r$ , not as a volumetric density.*

**Definition 16.2** (Quadratic geometric recruitment functional). *In the linear-response regime around an unloaded substrate, a scalar recruitment load field  $\chi$  sourced by a mass-energy distinction  $\mathcal{J}$  is represented by*

$$E_{\text{rec}}[\chi] = \frac{1}{2} \int_{\mathcal{M}} \mathcal{K}^{ij} \partial_i \chi \partial_j \chi d^3x - \int_{\mathcal{M}} \mathcal{J} \chi d^3x + O(\chi^3),$$

where  $\mathcal{K}^{ij}$  is the unloaded spatial comparison stiffness. The cubic and higher terms encode nonlinear self-coupling and are not fixed in this draft.

**Assumption 16.3** (Isotropic exterior kernel). *For a static, isolated, spherically symmetric source in the exterior linear-response regime, the unloaded spatial comparison stiffness is isotropic:*

$$\mathcal{K}^{ij} = K_s \delta^{ij}.$$

**Proposition 16.4** (Exterior point-source profile in the linear-response regime). *Under the quadratic recruitment functional and isotropic exterior kernel, a point source  $\mathcal{J}(x) = \mu M \delta^{(3)}(x)$  has an exterior equilibrium profile*

$$\chi_{\text{rec}}(r) = \frac{\mu M}{4\pi K_s r}$$

up to the normalization convention for  $\mu$ .

*Proof.* Varying  $E_{\text{rec}}$  gives the Euler-Lagrange equation

$$-\partial_i (\mathcal{K}^{ij} \partial_j \chi) = \mathcal{J}.$$

With  $\mathcal{K}^{ij} = K_s \delta^{ij}$ , this becomes

$$-K_s \nabla^2 \chi = \mu M \delta^{(3)}(x).$$

Using  $\nabla^2(1/r) = -4\pi \delta^{(3)}(x)$ , the exterior solution is

$$\chi_{\text{rec}}(r) = \frac{\mu M}{4\pi K_s r},$$

with any additive constant fixed by the boundary condition  $\chi \rightarrow 0$  as  $r \rightarrow \infty$ . □

*Remark 16.5* (What this does and does not close). Proposition 16.4 closes only the Green's-function step once the quadratic isotropic kernel is assumed. It does not derive the kernel from APF primitives, does not derive the source normalization  $\mu$ , and does not derive Newton's constant. It is nevertheless useful: it shows that the required  $1/r$  exterior profile is the natural equilibrium profile of the simplest geometric recruitment functional.

## 17. Cost-curvature response

**Principle 17.1** (Linear cost-curvature response). *In the exterior weak-load regime, the time-comparison cost-curvature responds linearly to the effective accumulated geometric recruitment load:*

$$K_t(x) = K_{t0}(1 - \alpha \chi_{\text{rec}}(x)),$$

where  $K_{t0}$  is the unloaded time-comparison stiffness and  $\alpha$  is the coupling between committed load and residual time-comparison stiffness.

The sign is physical: substrate already committed to carrying recruitment load has reduced residual stiffness for additional identity-carrying time comparisons. In represented geometry, this reduced stiffness appears as slower local proper-time accumulation relative to the unloaded coordinate normalization. In substrate-position-axis terms (Paper 0 §sec:substrate\_position\_axis), the sign is the substrate's middle-regime saturation-direction physics at the geometric-substrate scale: the closer the substrate is to its saturation regime — the more committed its capacity is to existing recruitment load — the less residual middle-regime capacity remains for additional comparison continuations. The Bekenstein-bound saturation-regime cardinal datum (Paper 6 Supplement v2.10 §sec:Lambda\_subspace,  $S_{\text{dS}} = 61 \log 102$ ) is the global-scale saturation limit of the same response. *Boundary 17.2* (Status of the response law). The response law is a structural commitment in this draft. A closed derivation would require a nonlinear geometric recruitment functional, expansion around the loaded equilibrium, and proof that the leading correction to  $K_{t0}$  is exactly  $-\alpha\chi_{\text{rec}}$ . That is one of the main open tasks of this paper.

## 17.1 Minimal perturbative response model

The following lemma does not close the response law from APF primitives. It formalizes the exact kind of load-probe coupling that would make the law unavoidable in a weak-load effective theory.

**Definition 17.3** (Time-comparison probe). *Let  $u$  be a small normalized probe field representing an infinitesimal rest-frame identity-comparison tick. In the unloaded substrate its leading quadratic cost is*

$$E_t[u] = \frac{1}{2} \int K_{t0} u^2 d\lambda,$$

where  $d\lambda$  is the local comparison-volume element appropriate to the clock sector.

**Assumption 17.4** (Bilinear load-probe coupling). *In the weak-load regime, the leading coupling between accumulated geometric recruitment load  $\chi$  and the time-comparison probe  $u$  is*

$$E_t[u; \chi] = \frac{1}{2} \int (K_{t0} - \lambda\chi(x)) u(x)^2 d\lambda + O(u^3, \chi^2 u^2),$$

with  $\lambda > 0$ . The minus sign states residual-stiffness depletion: already committed geometric load reduces available stiffness for an additional identity-comparison tick.

**Proposition 17.5** (Linear response from bilinear load-probe coupling). *Under the bilinear load-probe coupling, the local time-comparison stiffness at fixed load is*

$$K_t(x) = K_{t0} - \lambda\chi(x) + O(\chi^2),$$

or equivalently

$$K_t(x) = K_{t0}(1 - \alpha\chi(x)) + O(\chi^2), \quad \alpha := \lambda/K_{t0}.$$

*Proof.* The local stiffness is the second variation of the probe cost with respect to the probe amplitude at the loaded equilibrium:

$$K_t(x) = \left. \frac{\delta^2 E_t[u; \chi]}{\delta u(x)^2} \right|_{u=0}.$$

Taking the second variation of the displayed effective cost gives  $K_{t0} - \lambda\chi(x)$  to first order in  $\chi$ . Dividing by  $K_{t0}$  gives the normalized coefficient  $\alpha = \lambda/K_{t0}$ .  $\square$

*Audit Note 17.6* (What the response lemma buys). Proposition 17.5 converts the response law from a naked ansatz into a precise perturbative target. The remaining open work is now sharper: derive the negative bilinear load-probe coupling from the nonlinear geometric substrate functional, rather than merely assert the final linear response.

**Definition 17.7** (Newtonian calibration). *In physical units, define the exterior point-mass calibration by*

$$\alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2}.$$

*Equivalently, the combination  $\alpha\mu/(4\pi K_s)$  is fixed to  $2G/c^2$ . In this draft,  $G$  is a substrate-specific dimensional input, not an internally derived number.*

**Corollary 17.8** (Schwarzschild time factor as calibrated cost curvature). *Under the point-source profile and Newtonian calibration,*

$$K_t(r) = K_{t0} \left( 1 - \frac{2GM}{rc^2} \right)$$

*in the exterior linear-response regime.*

## 18. Proper time as normalized signed clock-sector comparison length

**Definition 18.1** (Proper-time dictionary). *In the represented metric branch, the proper-time element for a static identity-preserving held continuation is the normalized signed clock-sector comparison length assigned by the represented causal kernel:*

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt.$$

*This dictionary treats  $dt$  as the unloaded coordinate comparison parameter and  $d\tau$  as the local identity-continuation clock tick.*

**Remark 18.2** (Why the square root). The local line element is quadratic in infinitesimal comparison displacement. Cost-curvature multiplies the quadratic form; the represented cost length is therefore the square root of the curvature-normalized quadratic element. This is a representation dictionary, not yet a derivation from a fully closed substrate action.

**Boundary 18.3** (Rest-frame bridge, not a signature derivation). The proper-time dictionary is a calibration of the signed comparison kernel on static timelike probes. It does not follow merely from the positive pullback response norm, and it does not identify general implementation cost with Lorentzian interval. Null directions can have zero signed interval while carrying nonzero physical maintenance or realization cost.

**Definition 18.4** (Clock sector). *The clock sector of a represented geometry is the part of the comparison-cost kernel that controls static rest-frame identity continuations. In a static coordinate representation it is encoded by a lapse factor*

$$N(x) := \frac{d\tau(x)}{dt}.$$

*The redshift calculation constrains this sector only.*

**Definition 18.5** (Spatial comparison sector). *The spatial comparison sector is the part of the comparison-cost kernel that controls endpoint comparison at fixed clock label. In a weak-field isotropic representation one may write, schematically,*

$$ds^2 = -N(x)^2 c^2 dt^2 + S(x)^2 d\mathbf{x}^2,$$

*where redshift fixes the lapse factor  $N$  but not the spatial scale factor  $S$ .*

**Proposition 18.6** (Redshift does not fix full geometry). *Any two static represented geometries with the same lapse factor  $N(r) = d\tau(r)/dt$  give the same static gravitational redshift between radii, even if their spatial comparison sectors differ. Therefore a redshift recovery alone cannot establish full GR.*

*Proof.* For static emitter and receiver, local frequency is inverse local proper period. The same coordinate period is compared against the two local lapse factors, so

$$\frac{\omega_A}{\omega_B} = \frac{N(r_B)}{N(r_A)}.$$

No spatial comparison factor enters this expression. Hence all represented metrics with the same lapse produce the same redshift.  $\square$

**Proposition 18.7** (Leading-order redshift recovery from cost-curvature response). *Consider a static, spherically symmetric recruitment load generated by a point mass  $M$ , in the exterior linear-response regime. Suppose*

$$K_t(r) = K_{t0}(1 - \alpha\chi_{\text{rec}}(r)), \quad \alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2},$$

*and suppose proper time is represented by normalized identity-continuation cost:*

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt.$$

*Then*

$$d\tau(r) = \sqrt{1 - \frac{2GM}{rc^2}} dt.$$

*Consequently, for a photon emitted by a static observer at  $r_B$  and received by a static observer at  $r_A > r_B$ ,*

$$\frac{\omega_A}{\omega_B} = \sqrt{\frac{1 - 2GM/r_Bc^2}{1 - 2GM/r_Ac^2}}.$$

*In the weak-field limit,*

$$\frac{\omega_A}{\omega_B} \approx 1 - \frac{GM}{c^2} \left( \frac{1}{r_B} - \frac{1}{r_A} \right).$$

*Proof.* The first displayed formula follows immediately from substituting the calibrated recruitment load into the cost-curvature response law and then applying the proper-time dictionary. For a static emitter and receiver, the same coordinate-period label is compared against two different local proper-time rates. Local frequency is inverse local proper period, hence

$$\omega_A = \omega_B \frac{d\tau(r_B)}{d\tau(r_A)}.$$

Substitution gives the redshift ratio. Expanding  $\sqrt{1 - 2GM/rc^2}$  to first order gives the weak-field expression.  $\square$

*Boundary 18.8* (Boundary of the result). Proposition 18.7 is a conditional recovery, not a closed derivation of GR. It assumes: (i) the linear cost-curvature response, (ii) the exterior point-source recruitment profile, (iii) the proper-time-as-normalized-cost dictionary, and (iv)  $G$  as the substrate coupling in physical units. Deriving these from substrate primitives, extending the result beyond the exterior linear-response regime, and recovering the full Einstein equations are open.

## 19. No-smuggling audit for the redshift recovery

The redshift recovery is useful only if its inputs are named without disguise. The calculation has four layers.

| Layer                  | Status           | Content                                                                                                                                                            |
|------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuation grammar   | Imported         | The judgment $\Gamma \Vdash_C D \rightsquigarrow E$ and the interpretation of geometry as comparison-continuation cost are imported from Paper 10.                 |
| Exterior profile       | Cond./proved     | The $1/r$ profile is proved here only after assuming an isotropic quadratic exterior kernel.                                                                       |
| Clock response         | Assumed          | $K_t = K_{t0}(1 - \alpha\chi)$ is a linear-response commitment. Its sign and linearity are physically motivated but not derived from a nonlinear substrate action. |
| Unit calibration       | Input            | The equality $\alpha\chi = 2GM/rc^2$ imports the physical coupling $G$ . The calculation recovers the form of the redshift, not the numerical value of $G$ .       |
| Proper-time dictionary | Representational | $d\tau = \sqrt{K_t/K_{t0}}dt$ is the metric-regime dictionary between normalized identity-continuation cost and clock time.                                        |

*Audit Note 19.1* (What is actually earned). The calculation earns the following claim: if a point mass produces a potential-like recruitment load, if time-comparison stiffness is reduced linearly by that load, and if proper time is normalized cost length, then the static clock-redshift sector has the Schwarzschild form at leading exterior order. It does not earn the spatial metric, the equivalence principle, nonlinear field equations, or a derived value of  $G$ .

*Audit Note 19.2* (What would close the result). To turn the recovery into a derivation, one must derive the isotropic Green’s-function kernel, the universal response law, the proper-time dictionary, and the physical coupling normalization from APF substrate primitives. These are not cosmetic gaps; they are the main mathematical work of Paper 9.

## 20. Newtonian acceleration as a cost-gradient effect

The redshift calculation recovers the clock sector. The next target is motion. In represented weak-field geometry, slow test bodies accelerate along gradients of the same potential that controls  $g_{tt}$ . Cost-theoretically, this should emerge because least-cost identity-carrying worldlines bend toward lower continuation cost.

**Definition 20.1** (Effective cost potential). *Define the effective cost potential  $\Phi_C$  by*

$$\frac{K_t(r)}{K_{t0}} = 1 + \frac{2\Phi_C(r)}{c^2}.$$

*For the calibrated point-source profile,*

$$\Phi_C(r) = -\frac{GM}{r}.$$

**Proposition 20.2** (Newtonian acceleration under the geodesic-cost dictionary). *Assume the metric-representation gate closes and slow test-body worldlines extremize normalized identity-continuation cost. In the weak-field, slow-velocity regime with*

$$\frac{K_t}{K_{t0}} = 1 + \frac{2\Phi_C}{c^2},$$

*the represented coordinate acceleration is*

$$\mathbf{a} = -\nabla\Phi_C.$$

*For  $\Phi_C = -GM/r$ , this gives the inverse-square law.*

*Proof sketch.* Under the geodesic-cost dictionary, the action for a slow test body is proportional to

$$\int d\tau \approx \int \left( 1 + \frac{\Phi_C}{c^2} - \frac{v^2}{2c^2} \right) dt$$

up to an irrelevant constant normalization and sign convention. Extremizing is equivalent to extremizing the ordinary weak-field Lagrangian  $L = v^2/2 - \Phi_C$ , yielding  $\mathbf{a} = -\nabla\Phi_C$ . The step is conditional on the metric and geodesic-cost dictionaries.  $\square$

*Remark 20.3* (Why this matters). The same scalar cost load controls both clock rate and slow-body acceleration. This is the beginning of an equivalence-principle-like universality claim, but it is not yet a derivation of the equivalence principle. To close that claim, the response law must be shown universal for all admissible rest-frame identity continuations, independent of matter composition.

## 21. Weak-field two-sector kernel and the missing spatial response

The clock-sector recovery should be tested against the standard weak-field hierarchy. The redshift calculation gives a lapse. A gravitational theory also needs the spatial comparison response. The next formal object is therefore not a scalar clock stiffness alone, but a two-sector normalized cost kernel.

**Definition 21.1** (Normalized two-sector weak-field cost kernel). *In a static, isotropic, weak-field represented regime, write the normalized comparison-cost line element as*

$$ds_C^2 = -\Theta_t(r)c^2dt^2 + \Theta_s(r)d\mathbf{x}^2 + O(c^{-4}),$$

*where*

$$\Theta_t(r) := \frac{K_t(r)}{K_{t0}}, \quad \Theta_s(r) := \frac{K_s(r)}{K_{s0}}.$$

*The redshift result fixes  $\Theta_t$ . It does not fix  $\Theta_s$ .*

**Definition 21.2** (Weak-field cost potential). *For a static exterior point source, define the cost potential  $\Phi_C$  by*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}).$$

*Under the Newtonian calibration of the scalar recovery,*

$$\Phi_C(r) = -\frac{GM}{r}.$$



**Definition 21.3** (Spatial response parameter). *The weak-field spatial comparison response is defined by*

$$\Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

*The dimensionless coefficient  $\gamma_C$  is the APF/cost-theoretic counterpart of the standard PPN parameter  $\gamma_{\text{PPN}}$ . General relativity has  $\gamma_C = 1$ .*

**Proposition 21.4** (Clock observables fix only the lapse). *Gravitational time dilation, static redshift, and Newtonian slow-body acceleration determine the leading clock potential  $\Phi_C$  but do not determine the spatial-response coefficient  $\gamma_C$ .*

*Proof.* The redshift and time-dilation formulae depend only on the lapse factor  $\Theta_t^{1/2}$ . The Newtonian slow-motion limit follows from the same lapse through the weak-field particle action. The spatial factor  $\Theta_s$  enters null propagation and spatial-curvature observables at the same post-Newtonian order, but it does not enter the static redshift ratio and does not affect the lowest-order slow-velocity acceleration law. Therefore the clock sector fixes  $\Phi_C$  and leaves  $\gamma_C$  free.  $\square$

**Gate 21.5** (Spatial-response / PPN- $\gamma$  gate). *To recover the full leading weak-field Schwarzschild sector, the geometric substrate must determine the spatial comparison response*

$$\Theta_s(r) = 1 - 2\frac{\Phi_C(r)}{c^2} + O(c^{-4}),$$

*that is,*

$$\gamma_C = \gamma_{\text{PPN}} = 1.$$

*Until this gate closes, the paper has recovered the clock sector and Newtonian limit, but not light bending, Shapiro delay, or the full first-post-Newtonian exterior metric.*

**Gate 21.6** (Reciprocal clock-ruler response). *The spatial-response gate can be sharpened into a candidate closure principle. In the exterior weak-field scalar-load regime, the time-comparison and spatial-comparison sectors are required to be reciprocal to first post-Newtonian order:*

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}),$$

*or equivalently*

$$\Theta_s(r) = \Theta_t(r)^{-1} + O(c^{-4}).$$

*Physical reading: clock comparison and ruler comparison are conjugate faces of the same finite-coordination substrate. A load that reduces the residual rate of identity-carrying clock continuation must increase the cost of spatial endpoint comparison by the inverse factor if the same local null coordination boundary is to be used by all admissible probes. This is a gate, not a theorem: it must ultimately be derived from the microscopic comparison functional.*

**Proposition 21.7** (Reciprocal response implies the PPN spatial coefficient). *Assume the two-sector weak-field form*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}), \quad \Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

*If the reciprocal clock-ruler gate holds, then*

$$\gamma_C = 1.$$

*Proof.* Multiplying the two sector factors gives

$$\begin{aligned}\Theta_s\Theta_t &= \left(1 - 2\gamma_C \frac{\Phi_C}{c^2}\right) \left(1 + 2\frac{\Phi_C}{c^2}\right) + O(c^{-4}) \\ &= 1 + 2(1 - \gamma_C) \frac{\Phi_C}{c^2} + O(c^{-4}).\end{aligned}$$

The reciprocal gate requires this product to be unity through order  $c^{-2}$ . For a nonzero exterior potential this forces  $1 - \gamma_C = 0$ , hence  $\gamma_C = 1$ .  $\square$

*Audit Note 21.8* (Why naive same-sign scalar depletion fails). A tempting first guess is that every comparison stiffness is depleted in the same scalar way:

$$\Theta_s(r) = \Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}).$$

Compared with the PPN form  $\Theta_s = 1 - 2\gamma_C\Phi_C/c^2$ , this gives  $\gamma_C = -1$ . Thus redshift would still be recovered from the clock sector, but the spatial-curvature contribution needed for the standard two-sector weak-field observables would have the wrong coefficient. The geometric substrate cannot be modeled as uniform same-sign depletion of every comparison channel. The spatial sector must be conjugate, tensorial, or otherwise constrained; this is precisely the content of Gate 21.6 and Gate 21.5.

## 21.1 Candidate derivation route: clock-ruler comparison-cell conservation

The reciprocal clock-ruler condition should not remain a slogan. The following definitions isolate a sharper operational certificate. The certificate is still a gate, but it is weaker and more testable than simply assuming  $\gamma_C = 1$ : it says that, direction by direction, the elementary comparison cell built from one clock tick and one spatial endpoint ruler is preserved by a static scalar load to first post-Newtonian order.

**Definition 21.9** (Directionwise clock-ruler comparison cell). *Fix a represented spatial comparison direction  $e$  in the static weak-field regime. The normalized clock-ruler comparison cell in that direction is*

$$\mathcal{A}_e(r) := \left(\frac{d\tau(r)}{dt}\right) \left(\frac{d\ell_e(r)}{dx_e}\right) = \sqrt{\Theta_t(r)\Theta_{s,e}(r)},$$

where  $d\ell_e = \sqrt{\Theta_{s,e}} dx_e$  is the represented spatial comparison length in direction  $e$ . In the isotropic weak-field sector,  $\Theta_{s,e} = \Theta_s$  for all spatial directions.

**Gate 21.10** (Directionwise comparison-cell conservation). *In an exterior, static, scalar-load regime with no shear, torsion, anisotropic substrate recruitment, or composition-dependent comparison channel, the elementary clock-ruler comparison cell is conserved to first post-Newtonian order:*

$$\mathcal{A}_e(r) = 1 + O(c^{-4})$$

for every represented spatial direction  $e$ .

*Physical reading: a scalar geometric load may redistribute comparison capacity between clock-rate continuation and endpoint-ruler continuation, but it does not create or destroy the local directionwise comparison cell. Clock slowing and ruler stretching are conjugate responses of the same finite-coordination substrate.*

**Proposition 21.11** (Cell conservation implies reciprocal clock-ruler response). *In the static isotropic two-sector weak-field kernel, directionwise comparison-cell conservation implies*

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}).$$

Consequently, under the PPN expansion of Proposition 21.7, it implies  $\gamma_C = 1$ .

*Proof.* In the isotropic sector,  $\mathcal{A}_e = \sqrt{\Theta_t\Theta_s}$  is independent of  $e$ . The cell-conservation gate gives  $\sqrt{\Theta_t\Theta_s} = 1 + O(c^{-4})$ . Squaring gives  $\Theta_t\Theta_s = 1 + O(c^{-4})$ . The implication  $\gamma_C = 1$  then follows from Proposition 21.7.  $\square$

*Audit Note 21.12* (What the cell-conservation route does not prove). Cell conservation is not the local speed of light, coordinate invariance, or redshift in disguise. Local null measurements give a universal measured speed once the metric representation is assumed; they do not by themselves fix  $\Theta_s$ . The new content is the claim that a static scalar recruitment load preserves the directionwise clock-ruler comparison cell at order  $c^{-2}$ . This can fail if the load creates anisotropic shear, if clock and ruler comparisons draw on independent substrate reservoirs, if matter species couple to different comparison kernels, or if the geometric load changes the local comparison-cell budget rather than redistributing it. These are empirical and mathematical failure modes, not notational details.

## 21.2 Sharper candidate route: canonical comparison pairing

The cell-conservation gate can itself be decomposed. A comparison between events has two faces: a primal clock-ruler displacement and a dual frequency-wavevector probe. The dimensionless action/phase pairing between these faces is the invariant bookkeeping object. This subsection isolates the extra condition needed to make that pairing imply reciprocal clock-ruler response.

**Definition 21.13** (Local canonical comparison pair). *Fix a spatial comparison direction  $e$  in a represented local frame. The primal comparison variables are*

$$q_e = (\tau, \ell_e),$$

where  $\tau$  is clock continuation and  $\ell_e$  is endpoint-ruler continuation in direction  $e$ . The dual probe variables are

$$p_e = (\omega, k_e),$$

where  $\omega$  is frequency cost and  $k_e$  is directional wave-number cost. The local comparison one-form is

$$\vartheta_e = -\omega d\tau + k_e d\ell_e.$$

Equivalently,  $\vartheta_e/\hbar$  is the local dimensionless phase increment of a probe used to test the comparison.

**Gate 21.14** (Canonical load transport). *A static scalar recruitment load acts canonically on the local comparison pair if, direction by direction, its transport map preserves the comparison one-form up to terms of order  $c^{-4}$ :*

$$T_\chi^* \vartheta_e = \vartheta_e + O(c^{-4}).$$

For a diagonal weak-field response this means that if

$$d\tau \mapsto a_t d\tau, \quad d\ell_e \mapsto a_s d\ell_e,$$

then the dual variables transform inversely,

$$\omega \mapsto a_t^{-1} \omega, \quad k_e \mapsto a_s^{-1} k_e,$$

so that the phase/action pairing is not changed by relabeling the same comparison through the loaded substrate.

*Audit Note 21.15* (Canonical pairing alone is not enough). Gate 21.14 preserves the action/phase pairing, but it does not by itself force  $a_t a_s = 1$ . Any nonzero product  $a_t a_s$  can be paired with inverse dual scaling and still preserve  $\vartheta_e$ . Therefore the spatial coefficient  $\gamma_C = 1$  requires one additional physical statement: the scalar load may exchange clock and ruler comparison capacity, but it may not dilate the elementary primal clock-ruler comparison cell at order  $c^{-2}$ .

**Gate 21.16** (Trace-free clock-ruler load transport). *In the exterior static scalar-load regime, after removing coordinate relabelings and anisotropic shear, the primal action of the load on each local clock-ruler comparison plane is trace-free to first post-Newtonian order. Equivalently, there exists a scalar load parameter  $\sigma(r) = O(c^{-2})$  such that*

$$d\tau \mapsto e^{\sigma(r)} d\tau, \quad d\ell_e \mapsto e^{-\sigma(r)} d\ell_e$$

for every spatial direction  $e$ , up to  $O(c^{-4})$ .

*Physical reading: a static scalar geometric load does not create an independent local comparison-cell resource. It transfers comparison capacity between conjugate clock and ruler channels. The trace part would represent creation or destruction of the directionwise comparison cell; the trace-free part represents redistribution.*

**Proposition 21.17** (Trace-free transport implies cell conservation). *Assume the represented two-sector weak-field kernel and trace-free clock-ruler load transport. Then directionwise clock-ruler comparison-cell conservation holds:*

$$\mathcal{A}_e(r) = 1 + O(c^{-4}).$$

Consequently,

$$\Theta_s(r)\Theta_t(r) = 1 + O(c^{-4}) \quad \text{and} \quad \gamma_C = 1.$$

*Proof.* By definition, the normalized clock factor is  $d\tau/dt = \sqrt{\Theta_t}$ , and the normalized ruler factor in direction  $e$  is  $d\ell_e/dx_e = \sqrt{\Theta_{s,e}}$ . Trace-free transport gives these factors as  $e^\sigma$  and  $e^{-\sigma}$ , up to  $O(c^{-4})$ . Hence

$$\mathcal{A}_e = \left(\frac{d\tau}{dt}\right) \left(\frac{d\ell_e}{dx_e}\right) = e^\sigma e^{-\sigma} + O(c^{-4}) = 1 + O(c^{-4}).$$

The remaining implications are exactly Propositions 21.7 and 21.11. □

**Corollary 21.18** (Nested spatial-gate chain). *The leading spatial coefficient can be closed by any of the following increasingly primitive certificates:*

$$\begin{aligned} \text{trace-free clock-ruler load transport} &\implies \text{directionwise cell conservation} \\ &\implies \Theta_s \Theta_t = 1 + O(c^{-4}) \\ &\implies \gamma_C = 1. \end{aligned}$$

*Thus the next mathematical target is sharper than  $\gamma_C = 1$ : derive trace-free transport from the geometric recruitment functional, or exhibit a countermodel showing that APF permits a trace component.*

### 21.3 Variational route: fixed-cell response and the trace obstruction

Trace-free transport should itself be generated by a local response principle, not merely imposed. The cleanest way to state the target is to decompose the diagonal clock-ruler load response into its trace and trace-free parts.

**Definition 21.19** (Logarithmic clock-ruler response variables). *Fix a represented spatial direction  $e$ . Write the local diagonal load response as*

$$\frac{d\tau}{dt} = a_t = e^{\xi_t}, \quad \frac{d\ell_e}{dx_e} = a_s = e^{\xi_s}.$$

*Define the trace and exchange variables*

$$u_e := \xi_t + \xi_s, \quad v_e := \xi_t - \xi_s.$$

*Then  $u_e = \log \mathcal{A}_e$  is the logarithmic clock-ruler cell dilation, while  $v_e$  is the trace-free exchange between clock and ruler comparison channels.*

**Gate 21.20** (Fixed-cell variational response). *In the exterior static scalar-load regime, after removing coordinate relabelings and shear, the local response of a directionwise clock-ruler comparison plane is governed to quadratic order by an effective potential of the form*

$$\mathcal{V}_e(u_e, v_e; \chi) = \frac{1}{2}A_e u_e^2 + \frac{1}{2}B_e(v_e - \beta_e \chi)^2 + O(|u_e|^3 + |v_e|^3 + \chi^2|u_e| + \chi^2|v_e|),$$

*with  $A_e, B_e > 0$ . The scalar load sources the exchange mode  $v_e$  but does not source the trace mode  $u_e$  at first order.*

*Physical reading: committed geometric recruitment may shift capacity between clock continuation and ruler comparison, but a scalar static load has no independent channel for creating or destroying the elementary clock-ruler comparison cell. Any first-order source term proportional to  $\chi u_e$  would be a new local comparison-resource reservoir and must be separately justified.*

**Proposition 21.21** (Fixed-cell variational response implies trace-free transport). *Assume the fixed-cell variational response gate and  $\chi = O(c^{-2})$ . The local equilibrium response satisfies*

$$u_e = O(\chi^2) = O(c^{-4}), \quad v_e = \beta_e \chi + O(\chi^2).$$

*Consequently*

$$\mathcal{A}_e = e^{u_e} = 1 + O(c^{-4}),$$

*so directionwise comparison-cell conservation holds. In the isotropic two-sector weak-field regime this implies*

$$\Theta_s \Theta_t = 1 + O(c^{-4}), \quad \gamma_C = 1.$$

*Proof.* Equilibrium requires  $\partial \mathcal{V}_e / \partial u_e = 0$  and  $\partial \mathcal{V}_e / \partial v_e = 0$ . To first order,

$$\frac{\partial \mathcal{V}_e}{\partial u_e} = A_e u_e + O(\chi^2), \quad \frac{\partial \mathcal{V}_e}{\partial v_e} = B_e(v_e - \beta_e \chi) + O(\chi^2).$$

Since  $A_e, B_e > 0$ , the equilibrium solution is

$$u_e = O(\chi^2), \quad v_e = \beta_e \chi + O(\chi^2).$$

Because  $\chi = O(c^{-2})$ , the trace is absent through order  $c^{-2}$ . Hence  $\mathcal{A}_e = e^{u_e} = 1 + O(c^{-4})$ . The final implications are Propositions 21.7 and 21.11.  $\square$

## 21.4 Operator form of the no-trace target

The two-variable potential above is the scalar normal form of a more general loaded-equilibrium problem. In the represented comparison-metric regime, let  $q_0$  be the unloaded local comparison kernel and let  $\mathcal{S}$  be the finite-dimensional gauge-fixed space of symmetric perturbations  $h$  at that point. Write

$$\text{tr}_{q_0} h := q_0^{ab} h_{ab}, \quad P_{\text{tr}} h := \frac{1}{m} (\text{tr}_{q_0} h) q_0, \quad P_{\text{TF}} := I - P_{\text{tr}},$$

where  $m = \dim \mathcal{M}$  in the represented regime.

**Assumption 21.22** (Gauge-fixed quadratic recruitment response). *Around  $q_0$ , the local recruitment functional has expansion*

$$\begin{aligned} \mathcal{E}_{\text{rec}}[q_0 + h, \chi] &= \mathcal{E}_0 + \frac{1}{2} \langle h, \mathcal{K}h \rangle - \chi \langle \mathcal{J}, h \rangle \\ &\quad + O(\|h\|^3 + \chi \|h\|^2 + \chi^2 \|h\|), \end{aligned}$$

where  $\mathcal{K} : \mathcal{S} \rightarrow \mathcal{S}$  is self-adjoint and invertible after coordinate-gauge zero modes are removed, and  $\mathcal{J} \in \mathcal{S}$  is the scalar-load source.

**Theorem 21.23** (Linear response and the exact no-trace operator). *Under Assumption 21.22, the equilibrium perturbation is*

$$h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2).$$

*Its first-order comparison-volume variation is*

$$\delta_\chi \log \sqrt{|\det(q_0 + h_\chi)|} = \frac{\chi}{2} \text{tr}_{q_0}(\mathcal{K}^{-1} \mathcal{J}) + O(\chi^2).$$

*Consequently the no-trace gate is exactly*

$$\boxed{P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0}.$$

*Proof.* The stationary equation is

$$\mathcal{K}h_\chi - \chi \mathcal{J} + O(\|h_\chi\|^2 + \chi \|h_\chi\| + \chi^2) = 0.$$

Invertibility and the implicit-function expansion give  $h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2)$ . The determinant variation formula gives

$$\log \sqrt{|\det(q_0 + h)|} = \log \sqrt{|\det q_0|} + \frac{1}{2} \text{tr}_{q_0} h + O(\|h\|^2).$$

Substitution proves the volume formula. Vanishing of the trace component is equivalent to  $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ .  $\square$

**Corollary 21.24** (Source-selection route). *If*

$$P_{\text{tr}} \mathcal{J} = 0$$

*and  $\mathcal{K}$  preserves the trace-free subspace, equivalently*

$$P_{\text{tr}} \mathcal{K} P_{\text{TF}} = 0 = P_{\text{TF}} \mathcal{K} P_{\text{tr}},$$

*then  $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$  and the linear response is trace-free.*

**Corollary 21.25** (Constrained fixed-volume route). *If the admissible variation space is constrained by*

$$\mathrm{tr}_{q_0} h = 0$$

*through a Lagrange multiplier or a primitive fixed-comparison-capacity condition, then the equilibrium response is trace-free by construction. The remaining physical obligation is to derive the constraint from admissibility rather than impose it as a coordinate convention.*

*Audit Note 21.26* (Relation to the scalar potential). In a clock–ruler plane,  $u_e$  is the trace coordinate and  $v_e$  the trace-free exchange coordinate. The condition that no term  $\chi u_e$  appear is the two-dimensional source-selection statement  $P_{\mathrm{tr}} \mathcal{J} = 0$ . The stronger invariant target is  $P_{\mathrm{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ , because a trace-free source can still generate trace if the Hessian mixes the trace and trace-free sectors.

**Definition 21.27** (Trace-obstruction coefficient). *If the local response potential contains a first-order trace source,*

$$\mathcal{V}_e(u_e, v_e; \chi) = \frac{1}{2} A_e u_e^2 + \frac{1}{2} B_e (v_e - \beta_e \chi)^2 - \lambda_e \chi u_e + O(3),$$

*then the equilibrium trace is*

$$u_e = \eta_e \chi + O(\chi^2), \quad \eta_e := \lambda_e / A_e.$$

*The coefficient  $\eta_e$  is the first-order trace obstruction.*

**Proposition 21.28** (Trace obstruction is the PPN- $\gamma$  obstruction). *In the isotropic two-sector weak-field regime, suppose the logarithmic cell dilation has the expansion*

$$u(r) = \zeta_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

*Then*

$$\gamma_C = 1 - \zeta_C.$$

*Thus the GR spatial coefficient  $\gamma_C = 1$  is equivalent to vanishing first-order trace dilation  $\zeta_C = 0$ .*

*Proof.* By definition,

$$e^{2u} = \Theta_t \Theta_s.$$

Using the two-sector weak-field expansion,

$$\begin{aligned} \Theta_t \Theta_s &= \left(1 + 2 \frac{\Phi_C}{c^2}\right) \left(1 - 2\gamma_C \frac{\Phi_C}{c^2}\right) + O(c^{-4}) \\ &= 1 + 2(1 - \gamma_C) \frac{\Phi_C}{c^2} + O(c^{-4}). \end{aligned}$$

Meanwhile  $e^{2u} = 1 + 2\zeta_C \Phi_C / c^2 + O(c^{-4})$ . Equating coefficients gives  $\zeta_C = 1 - \gamma_C$ , hence  $\gamma_C = 1 - \zeta_C$ .  $\square$

*Audit Note 21.29* (The remaining obstruction). The spatial gate has now been reduced to a concrete question: does static scalar geometric recruitment source the trace mode  $u_e$  of the local clock-ruler response? If not, fixed-cell variational response gives trace-free transport and closes  $\gamma_C = 1$ . If yes, the sourced trace coefficient  $\zeta_C$  shifts the PPN spatial coefficient by  $\gamma_C = 1 - \zeta_C$ . The obstruction is no longer vague. It is the existence of a first-order  $\chi u_e$  term in the effective local response potential.

## 21.5 Generator form: the obstruction is a $GL(2)$ trace

The variational statement above is useful, but the same obstruction can be stated more invariantly. A weak static load acts locally on each clock-ruler comparison plane by a linear response map. The determinant of this map is the comparison-cell factor. The trace of its infinitesimal generator is therefore exactly the first-order cell-dilation obstruction.

**Definition 21.30** (Clock-ruler response map). *Fix a represented spatial direction  $e$ . After removing coordinate relabelings and shear, write the loaded clock-ruler increments as*

$$\begin{pmatrix} d\tau \\ d\ell_e \end{pmatrix} = R_e(\hat{\chi}) \begin{pmatrix} d\tau_0 \\ d\ell_{e0} \end{pmatrix}, \quad \hat{\chi}(r) := \frac{\Phi_C(r)}{c^2} = O(c^{-2}),$$

where  $R_e(0) = I$  and  $R_e(\hat{\chi}) \in GL^+(2, \mathbb{R})$  in the represented smooth regime. Its infinitesimal generator is

$$L_e := \left. \frac{dR_e}{d\hat{\chi}} \right|_0, \quad R_e(\hat{\chi}) = I + \hat{\chi}L_e + O(\hat{\chi}^2).$$

The trace component

$$\theta_e := \text{tr } L_e$$

is the infinitesimal comparison-cell dilation rate. The traceless part  $L_e - \frac{1}{2}\theta_e I$  is the exchange/shear part of the clock-ruler response.

**Proposition 21.31** (Generator trace equals trace obstruction). *In the diagonal isotropic weak-field sector, the logarithmic cell dilation satisfies*

$$u_e(r) = \theta_e \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

Consequently

$$\zeta_C = \theta_e, \quad \gamma_C = 1 - \theta_e.$$

Thus the GR spatial coefficient  $\gamma_C = 1$  is equivalent, at this order, to the infinitesimal load generator being traceless:

$$\text{tr } L_e = 0.$$

*Proof.* The comparison-cell factor is

$$\mathcal{A}_e = \det R_e(\hat{\chi}) = 1 + \hat{\chi} \text{tr } L_e + O(\hat{\chi}^2).$$

Since  $u_e = \log \mathcal{A}_e$ , we obtain

$$u_e = \hat{\chi} \text{tr } L_e + O(\hat{\chi}^2) = \theta_e \Phi_C / c^2 + O(c^{-4}).$$

Comparison with Proposition 21.28 gives  $\zeta_C = \theta_e$  and hence  $\gamma_C = 1 - \theta_e$ . □

## 21.6 Trace as comparison-volume variation

The  $GL(2)$  trace has an exact geometric interpretation once a represented signed comparison kernel is available on the local clock-ruler plane.

**Definition 21.32** (Loaded clock-ruler comparison kernel). *Let  $q_{e,0}$  be the unloaded nondegenerate comparison kernel on the clock-ruler plane. Represent the loaded kernel on the unloaded increment space by*

$$q_e(\hat{\chi}) := R_e(\hat{\chi})^* q_{e,0} = R_e(\hat{\chi})^\top q_{e,0} R_e(\hat{\chi}).$$



**Theorem 21.33** (Trace–volume identity). *For  $R_e(\hat{\chi}) = I + \hat{\chi}L_e + O(\hat{\chi}^2)$ ,*

$$\boxed{\left. \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_e(\hat{\chi})|} \right|_{\hat{\chi}=0} = \text{tr } L_e}.$$

*Thus the infinitesimal generator trace is precisely the first variation of the local comparison-volume element.*

*Proof.* The determinant transforms as

$$\det q_e(\hat{\chi}) = \det(R_e)^2 \det q_{e,0}.$$

Because  $R_e(0) = I$  and  $R_e \in GL^+(2, \mathbb{R})$  locally,

$$\log \sqrt{|\det q_e|} = \log \det R_e + \log \sqrt{|\det q_{e,0}|}.$$

Differentiating at zero and using  $\frac{d}{d\hat{\chi}} \log \det R_e|_0 = \text{tr } L_e$  proves the identity.  $\square$

**Corollary 21.34** (The PPN obstruction is the scalar volume mode). *After clock calibration,*

$$\left. \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_e(\hat{\chi})|} \right|_0 = 1 - \gamma_C.$$

*Hence  $\gamma_C = 1$  if and only if the static scalar load has no first-order local comparison-volume mode.*

## 21.7 Gauge-reduced scalar normal form

The generator formulation still permits too many apparent degrees of freedom. Some are physical, but some are only coordinate choice, anisotropic shear, or time-space mixing inappropriate to the static scalar exterior sector. The next reduction is to put the local response into normal form.

**Definition 21.35** (Gauge-reduced scalar response sector). *Fix a static, spherically symmetric exterior weak-field load. A local clock-ruler response map is in the gauge-reduced scalar sector when, after quotienting coordinate relabelings, anisotropic shear, and parity-forbidden time-space mixing, its infinitesimal generator is diagonal and direction-independent:*

$$L_{\text{sc}} = \begin{pmatrix} \lambda_t & 0 \\ 0 & \lambda_s \end{pmatrix} = \frac{\theta}{2} I + \sigma H, \quad H := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Here

$$\theta := \lambda_t + \lambda_s = \text{tr } L_{\text{sc}}, \quad \sigma := \frac{\lambda_t - \lambda_s}{2}.$$

*The trace  $\theta$  is the comparison-cell dilation mode; the exchange coefficient  $\sigma$  transfers comparison scale between clock and ruler channels without changing the cell determinant.*

**Proposition 21.36** (Clock calibration leaves only the trace obstruction). *In the gauge-reduced scalar sector, suppose the clock response is calibrated by the redshift recovery,*

$$\sqrt{\Theta_t} = 1 + \frac{\Phi_C}{c^2} + O(c^{-4}),$$

*and write the spatial response as*

$$\sqrt{\Theta_s} = 1 - \gamma_C \frac{\Phi_C}{c^2} + O(c^{-4}).$$

Then the normal-form generator has

$$\lambda_t = 1, \quad \lambda_s = -\gamma_C,$$

so that

$$\theta = 1 - \gamma_C, \quad \sigma = \frac{1 + \gamma_C}{2}.$$

Therefore, once the clock sector is calibrated, the entire first-order spatial-response problem is the trace problem:

$$\gamma_C = 1 \iff \theta = 0.$$

*Proof.* By Definition 21.30, the loaded increments are

$$d\tau = (1 + \hat{\chi}\lambda_t)d\tau_0 + O(\hat{\chi}^2), \quad d\ell = (1 + \hat{\chi}\lambda_s)d\ell_0 + O(\hat{\chi}^2),$$

with  $\hat{\chi} = \Phi_C/c^2$ . Matching these expressions to the displayed clock and ruler factors gives  $\lambda_t = 1$  and  $\lambda_s = -\gamma_C$ . The formulae for  $\theta$  and  $\sigma$  follow from Definition 21.35. Hence  $\gamma_C = 1$  iff  $\theta = 0$ .  $\square$

**Corollary 21.37** (Same-sign depletion failure in normal form). *A same-sign scalar depletion of clock and ruler increments has  $\lambda_t = \lambda_s = 1$ . With the clock sector calibrated, this implies*

$$\gamma_C = -1,$$

*so it reproduces the redshift sign but fails the spatial weak-field sector.*

*Proof.* If  $\lambda_t = 1$  and  $\lambda_s = 1$ , then  $-\gamma_C = 1$ , hence  $\gamma_C = -1$ . Equivalently, the trace is  $\theta = 2$ , so  $\gamma_C = 1 - \theta = -1$ .  $\square$

**Proposition 21.38** (Unique scalar closure compatible with calibrated weak-field GR). *Within the gauge-reduced scalar sector with the clock coefficient fixed by redshift, the following are equivalent to first post-Newtonian order:*

$$\begin{aligned} L_{\text{sc}} \in \mathfrak{sl}(2, \mathbb{R}) &\iff L_{\text{sc}} = H \\ &\iff R_{\text{sc}} \in SL(2, \mathbb{R}) + O(c^{-4}) \\ &\iff \Theta_s = \Theta_t^{-1} + O(c^{-4}) \\ &\iff \gamma_C = 1. \end{aligned}$$

*Thus the scalar weak-field closure is unique: the load must act as a trace-free clock-ruler exchange, not as a common dilation.*

*Proof.* Clock calibration gives  $\lambda_t = 1$ . Tracelessness gives  $\lambda_s = -1$ , hence  $L_{\text{sc}} = H$ . This is equivalent to  $\det R_{\text{sc}} = 1 + O(c^{-4})$ , since the determinant changes to first order by  $\text{tr } L_{\text{sc}}\hat{\chi}$ . The determinant condition is  $\Theta_s\Theta_t = 1 + O(c^{-4})$ , which is the same as  $\Theta_s = \Theta_t^{-1} + O(c^{-4})$ . By Proposition 21.36, this is equivalent to  $\gamma_C = 1$ .  $\square$

**Theorem 21.39** (Scalar weak-field closure package). *Assume the static isotropic exterior weak-field sector, the metric-representation gate, universal matter coupling to the same comparison kernel, and the clock calibration supplied by the redshift recovery. After coordinate gauge, anisotropic shear,*

and parity-forbidden time-space mixing are removed, the following statements are equivalent to first post-Newtonian order:

$$\begin{aligned}
\text{no scalar comparison-cell dilation} &\iff \text{tr } L_{\text{sc}} = 0 \\
&\iff L_{\text{sc}} \in \mathfrak{sl}(2, \mathbb{R}) \\
&\iff \det R_{\text{sc}} = 1 + O(c^{-4}) \\
&\iff \left. \frac{d}{d\hat{\chi}} \log \sqrt{|\det q_{\text{sc}}(\hat{\chi})|} \right|_0 = 0 \\
&\iff \Theta_s \Theta_t = 1 + O(c^{-4}) \\
&\iff \gamma_C = 1.
\end{aligned}$$

If these equivalent conditions hold, the conditional weak-field exterior recovery theorem applies. If they fail with trace  $\theta$ , then

$$\gamma_C = 1 - \theta,$$

so two-sector weak-field observations measure the scalar comparison-cell dilation directly.

*Proof.* Clock calibration gives  $\lambda_t = 1$ . In the gauge-reduced scalar sector, Proposition 21.36 gives  $\theta = 1 - \gamma_C$ , so  $\gamma_C = 1$  iff  $\theta = 0$ . The equivalence between vanishing trace, membership in  $\mathfrak{sl}(2, \mathbb{R})$ , first-order unimodularity of  $R_{\text{sc}}$ , and preservation of the normalized clock-ruler cell is the determinant identity for a first-order  $GL(2)$  response. In the isotropic two-sector kernel this determinant condition is  $\Theta_s \Theta_t = 1 + O(c^{-4})$ . The conditional weak-field recovery then follows from Theorem 21.49. If  $\theta \neq 0$ , Proposition 21.28 gives the shifted coefficient  $\gamma_C = 1 - \theta$ .  $\square$

*Audit Note 21.40* (What the scalar package reduces). The spatial problem is no longer an unconstrained metric-fitting problem. In the static scalar exterior sector, after gauge and shear are removed, redshift fixes  $\lambda_t$ . The only remaining first-order scalar ambiguity is  $\theta = \text{tr } L_{\text{sc}}$ . Proving the APF substrate has no independent scalar comparison-cell dilation is therefore enough to close the leading two-sector weak-field geometry.

**Gate 21.41** (Unimodular scalar load response). *Static scalar geometric recruitment acts on each local clock-ruler comparison plane by a unimodular response after coordinate gauge, anisotropic shear, and composition-dependent channels are removed:*

$$\det R_e(\hat{\chi}) = 1 + O(c^{-4}), \quad \text{equivalently} \quad L_e \in \mathfrak{sl}(2, \mathbb{R}).$$

*Physical reading: scalar recruitment may redistribute comparison capacity between clock and ruler channels, but it does not create or destroy the elementary local comparison cell. A failure of this gate is not a small technical issue; it is an additional scalar trace mode in the weak-field geometry.*

**Proposition 21.42** (Unimodular response closes the spatial coefficient). *Assume the two-sector weak-field kernel and the unimodular scalar load response gate. Then*

$$\Theta_s \Theta_t = 1 + O(c^{-4}), \quad \gamma_C = 1.$$

*Proof.* The determinant of the clock-ruler response map is the normalized comparison-cell factor:

$$\det R_e = \mathcal{A}_e = \left( \frac{d\tau}{dt} \right) \left( \frac{d\ell_e}{dx_e} \right) = \sqrt{\Theta_t \Theta_{s,e}}.$$

Unimodularity gives  $\mathcal{A}_e = 1 + O(c^{-4})$ , hence  $\Theta_t \Theta_{s,e} = 1 + O(c^{-4})$ . In the isotropic sector this is  $\Theta_t \Theta_s = 1 + O(c^{-4})$ , and Proposition 21.7 gives  $\gamma_C = 1$ .  $\square$

**Corollary 21.43** (Equivalence of the spatial closure targets). *In the isotropic weak-field regime, the following are equivalent to first post-Newtonian order:*

$$\begin{aligned}
\text{tr } L_e = 0 &\iff \det R_e = 1 + O(c^{-4}) \\
&\iff \mathcal{A}_e = 1 + O(c^{-4}) \\
&\iff \Theta_s \Theta_t = 1 + O(c^{-4}) \\
&\iff \gamma_C = 1.
\end{aligned}$$

*The spatial gate can therefore be pursued as any one of four equivalent problems: rule out the trace generator, prove unimodular response, prove comparison-cell conservation, or derive reciprocal clock-ruler response.*

*Audit Note 21.44* (Operational diagnostic for the trace mode). If weak-field tests determine an effective PPN coefficient  $\gamma_{\text{obs}}$ , the APF trace mode is read off as

$$\theta_e = \zeta_C = 1 - \gamma_{\text{obs}}.$$

Thus the observational role of light bending, Shapiro delay, and other two-sector probes is not merely to test GR. In the cost language, they bound or detect the first-order determinant of the local clock-ruler response map. Redshift alone cannot do this, because it probes only the clock projection.

**Proposition 21.45** (Nonzero trace is a scalar comparison mode). *Within the two-sector weak-field ansatz, a nonzero first-order trace generator  $\theta_e \neq 0$  is observationally equivalent, at this order, to adding an independent scalar comparison mode that changes the spatial PPN coefficient while leaving the redshift sector fixed.*

*Proof.* The redshift sector fixes  $\Theta_t = 1 + 2\Phi_C/c^2 + O(c^{-4})$ . A nonzero trace generator changes the product  $\Theta_t \Theta_s$  by

$$\Theta_t \Theta_s = 1 + 2\theta_e \frac{\Phi_C}{c^2} + O(c^{-4}),$$

which by Proposition 21.31 shifts  $\gamma_C$  to  $1 - \theta_e$ . Since  $\Theta_t$  is unchanged, the new degree of freedom is invisible to clock-sector redshift but visible to two-sector spatial observables. That is exactly the weak-field role of an additional scalar comparison mode.  $\square$

*Remark 21.46* (Why this is a strengthening). The variational obstruction can be expressed invariantly as a generator question: is scalar load transport  $GL(2)$  with nonzero trace, or  $SL(2)$  to first order? This is the cleanest current statement of the spatial-response problem.

**Corollary 21.47** (A sharper spatial-response target). *The weak-field spatial sector can be closed by proving the unimodular scalar load response gate Gate 21.41, or equivalently any of the targets listed in Corollary 21.43. One convenient proof chain is*

$$\begin{aligned}
\text{tr } L_e = 0 &\implies \det R_e = 1 + O(c^{-4}) \\
&\implies \Theta_s \Theta_t = 1 + O(c^{-4}) \\
&\implies \gamma_C = 1 \\
&\implies \text{leading two-sector weak-field GR recovery.}
\end{aligned}$$

*Thus the next target is no longer the bare coefficient  $\gamma_C$ , but the underlying no-trace / unimodular comparison-load certificate.*

**Assumption 21.48** (Two-sector weak-field response). *In the exterior weak-field regime generated by a static point source, the represented comparison kernel has*

$$\Theta_t(r) = 1 + \frac{2\Phi_C(r)}{c^2} + O(c^{-4}), \quad \Theta_s(r) = 1 - 2\gamma_C \frac{\Phi_C(r)}{c^2} + O(c^{-4}).$$

*The clock calculation gives the first relation conditionally. The second relation is the spatial-response target.*

**Theorem 21.49** (Conditional weak-field exterior recovery). *Assume the metric-representation gate, the exterior point-source profile, the Newtonian calibration  $\Phi_C = -GM/r$ , the proper-time dictionary, universal matter coupling to the same comparison kernel, and the spatial-response gate  $\gamma_C = 1$ . Then the represented exterior weak-field cost kernel is*

$$ds_C^2 = - \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 + \frac{2GM}{rc^2}\right) d\mathbf{x}^2 + O(c^{-4}),$$

*which is the first-post-Newtonian Schwarzschild exterior metric in isotropic weak-field form. Consequently the framework conditionally recovers, at leading weak-field order, gravitational redshift, clock-height tests, Newtonian acceleration, light bending, and Shapiro time delay.*

*Proof.* The clock-sector recovery and calibration give

$$\Theta_t(r) = 1 - \frac{2GM}{rc^2} + O(c^{-4}).$$

The spatial-response gate with  $\gamma_C = 1$  gives

$$\Theta_s(r) = 1 + \frac{2GM}{rc^2} + O(c^{-4}).$$

Substitution into Definition 21.1 gives the displayed metric kernel. In standard post-Newtonian bookkeeping, this pair of coefficients is exactly the weak-field Schwarzschild exterior through the leading order controlling the listed observables. The recovery is conditional because the metric-representation gate, universal matter coupling, and either the spatial-response gate or its sharper reciprocal clock-ruler route are not yet derived from substrate primitives.  $\square$

**Corollary 21.50** (No-trace scalar response is sufficient for the leading weak-field package). *Under the hypotheses of Theorem 21.49, the explicit spatial-response assumption  $\gamma_C = 1$  may be replaced by any one of the equivalent scalar-closure conditions in Theorem 21.39. In particular, if the gauge-reduced scalar load generator is traceless,*

$$\text{tr } L_{\text{sc}} = 0,$$

*then the conditional first-post-Newtonian exterior recovery follows.*

*Proof.* By Theorem 21.39, vanishing trace is equivalent to  $\gamma_C = 1$  in the static isotropic scalar exterior sector after clock calibration. Substituting this equivalent gate for the spatial-response assumption gives exactly the hypotheses of Theorem 21.49.  $\square$

**Corollary 21.51** (Observable dependency split). *Within the two-sector weak-field kernel, redshift and clock-height tests are clock-sector observables, while light bending and Shapiro delay are two-sector observables. Therefore any APF claim to recover leading weak-field GR must close both the clock-response gate and the spatial-response gate.*

| Observable              | Needs $\Theta_t$ | Needs $\Theta_s$              | Present status                                                              |
|-------------------------|------------------|-------------------------------|-----------------------------------------------------------------------------|
| Static redshift         | Yes              | No                            | Recovered conditionally from clock response.                                |
| Clock-height tests      | Yes              | No                            | Same gate as redshift, with local calibration.                              |
| Newtonian acceleration  | Yes              | No at leading slow-body order | Conditional on geodesic-cost dictionary and universal matter coupling.      |
| Light bending           | Yes              | Yes                           | Open until $\gamma_C = 1$ is derived.                                       |
| Shapiro delay           | Yes              | Yes                           | Open until null comparison uses the two-sector kernel with $\gamma_C = 1$ . |
| Perihelion precession   | Yes              | Yes plus next-order terms     | Partly first-PN; full recovery needs nonlinear corrections.                 |
| Frame dragging          | Needs $h_{0i}$   | Yes                           | Deferred to rotating-source tensor sector.                                  |
| Gravitational radiation | Tensor dynamics  | Tensor dynamics               | Deferred to dynamical self-coupled geometric substrate.                     |

*Audit Note 21.52* (No-smuggling table for the weak-field suite).

## 22. From response cost geometry to tensorial comparison response

The scalar calculation used one effective load  $\chi$  and one clock stiffness. The intrinsic construction now shows that a tensorial theory may carry two different quadratic objects:

$$g_{ab}^{\text{cost}} \quad \text{and} \quad q_{ab}^{\text{cmp}}.$$

The first, when it exists, is the positive metric induced by calibrated response effort. The second is the signed comparison kernel supplied by the causal/signature gate. Their relation is a bridge theorem, not an identity inherited from notation.

**Definition 22.1** (Tensorial signed comparison kernel). *A tensorial signed comparison kernel is a smooth nondegenerate field  $q_{ab}^{\text{cmp}}$  whose quadratic function*

$$Q_x^{\text{cmp}}(dx) = q_{ab}^{\text{cmp}}(x) dx^a dx^b$$

*represents the causal comparison structure. When the signature gate closes it has Lorentzian signature. The positive implementation effort of the same displacement is measured separately by  $g^{\text{cost}}$  or by the underlying Finsler gauge.*

**Definition 22.2** (Weak perturbation around unloaded comparison geometry). *In the represented weak-field regime, write*

$$q_{ab}^{\text{cmp}} = \eta_{ab} + h_{ab}, \quad |h_{ab}| \ll 1,$$

*where  $\eta_{ab}$  is the unloaded Lorentzian comparison kernel and  $h_{ab}$  the recruited comparison distortion.*

**Principle 22.3** (Universal comparison-load coupling). *Matter distinctions couple to the represented comparison geometry through the burden they impose on the substrate. In the tensorial regime this burden is summarized by a stress-energy functional  $T^{ab}$ , and the leading coupling has schematic form*

$$E_{\text{rec}}[h; T] = E_{\text{geom}}[h] - \lambda \int h_{ab} T^{ab} d^4x + O(h^2 T, h T^2).$$

*Universality of free fall requires the same comparison kernel to be read by all admissible matter sectors. The positive cost geometry must be compatible with this coupling through the anchor and cost-causal bridge.*

**Gate 22.4** (No composition dependence gate). *The equivalence-principle-like claim closes only if the recruitment load of a composite body depends on its total carried stress-energy distinction and not on microscopic composition except through already-accounted internal energy. All admissible clocks and test bodies must read the same signed comparison kernel and its calibrated cost response.*

**Proposition 22.5** (Redshift sector as the  $h_{00}$  projection). *In the weak static regime, redshift fixes*

$$h_{00}(r) \simeq \frac{2\Phi_C(r)}{c^2} = -\frac{2GM}{rc^2}$$

*up to the sign convention of the comparison kernel. It does not fix  $h_{ij}$ ,  $h_{0i}$ , or the positive cost metric  $g^{\text{cost}}$  away from the calibrated clock direction.*

*Proof.* Static redshift compares rest-frame time increments and therefore projects the signed comparison kernel onto its time-time component. Spatial, mixed, and independent cost-geometry components require additional probes and bridge laws.  $\square$

## 23. Translation route to the Paper 6 field equations

The Paper 6 route to GR is not to guess the Schwarzschild metric, but to characterize the admissible local cost functional for  $Q_{ab}$  after the represented smooth Lorentzian gates close. The expected structure is:

$$E_{\text{geom}}[Q] = \int_{\mathcal{M}} \left[ \text{local scalar made from } Q \text{ and its first/second derivatives} \right] d\mu_Q.$$

Paper 6 supplies this represented-geometry closure using H1–H3, A9, and Lovelock uniqueness. Papers 24/25 already discharge H1 and H2 (continuum emergence and locality of the  $\Delta_{\text{geo}}$  closure) at the recruitment level, so the continuum hypotheses needing translation reduce to H3 and the A9/Lovelock step. The Paper 9 challenge is narrower: translate those residual hypotheses into finite-continuability certificates and identify their weak-field scalar face.

**Gate 23.1** (Coordinate-relabeling invariance gate). *The cost assigned to a comparison-continuation configuration may not depend on arbitrary labels used for comparison sites. In the smooth represented regime, this becomes diffeomorphism-type invariance of the geometric cost functional.*

**Gate 23.2** (Second-order locality gate). *If the substrate cost is local and the represented field equations are to avoid additional higher-derivative propagating comparison modes, the continuum Euler-Lagrange equations should be second order in the metric kernel. This is the gate that points toward curvature scalar functionals rather than arbitrary higher-curvature actions.*

**Gate 23.3** (Bianchi/ledger gate). *The geometric side of the represented field equation must have an identically conserved ledger divergence so that local stress-energy continuation is compatible with the geometry equation. In GR this is the contracted Bianchi identity. In cost language it is the statement that relabeling comparison sites cannot create or destroy committed load.*

**Definition 23.4** (Geometric Euler ledger). *Let  $E_{\text{geom}}[Q]$  be a represented geometric cost functional. Its Euler ledger is the variational tensor density  $\mathcal{E}^{ab}$  defined by*

$$\delta E_{\text{geom}}[Q] = \int \mathcal{E}^{ab} \delta Q_{ab} d\mu_Q + \text{boundary terms.}$$

*A matter recruitment functional contributes a ledger  $\mathcal{T}^{ab}$  in the same variational channel.*

**Proposition 23.5** (Relabeling invariance implies a conserved geometric ledger). *If  $E_{\text{geom}}[Q]$  is invariant under compactly supported infinitesimal relabelings of comparison sites, then its Euler ledger satisfies a formal conservation identity*

$$\nabla_a \mathcal{E}^{ab} = 0$$

*in the represented smooth metric regime, up to the boundary conventions of the functional.*

*Proof sketch.* An infinitesimal relabeling generated by a vector field  $\xi^a$  changes the metric kernel by its Lie derivative,

$$\delta Q_{ab} = \mathcal{L}_\xi Q_{ab} = \nabla_a \xi_b + \nabla_b \xi_a.$$

Relabeling invariance gives  $\delta E_{\text{geom}} = 0$  for all compactly supported  $\xi$ . Substituting the variation and integrating by parts yields

$$0 = -2 \int (\nabla_a \mathcal{E}^{ab}) \xi_b d\mu_Q.$$

Since  $\xi$  is arbitrary,  $\nabla_a \mathcal{E}^{ab} = 0$ . The result is a represented-regime conservation identity; translating it back to finite continuations is a separate APF gate.  $\square$

**Corollary 23.6** (Matter compatibility condition). *If the represented field equation has the form*

$$\mathcal{E}^{ab} = \lambda \mathcal{T}^{ab},$$

*then relabeling invariance of the geometric ledger requires*

$$\nabla_a \mathcal{T}^{ab} = 0$$

*inside the same represented regime. In APF terms, local committed recruitment load must continue consistently with the comparison geometry that carries it.*

**Theorem 23.7** (Translation target: Einstein-Hilbert uniqueness, schematic). *If the represented geometric cost functional is local, smooth, coordinate-relabeling invariant, built only from the tensorial comparison kernel and its derivatives, yields second-order equations, and has a conserved ledger divergence compatible with universal stress-energy recruitment, then the leading continuum functional is of Einstein-Hilbert type,*

$$E_{\text{geom}}[Q] \propto \int (R[Q] - 2\Lambda) d\mu_Q,$$

*up to boundary terms, normalization, and higher-order corrections outside the gate.*

*Boundary 23.8* (Status of Theorem 23.7). This theorem is not reproved here. It is the Paper 9 translation target corresponding to the Paper 6 A9/Lovelock closure. Paper 6 supplies the represented-geometry proof; Paper 9 asks which finite-continuity certificates force the same continuum hypotheses and, in the weak-field scalar sector, which certificate forces the no-trace condition  $\theta = 0$ .



## 24. Nonlinear alignment: from response law to the Paper 6 closure

The leading-order theory uses a scalar accumulated load  $\chi$ . Paper 6 supplies the represented tensorial, nonlinear, self-coupled closure to the Einstein equations. The remaining Paper 9 problem is to show how the finite comparison-substrate response lifts into exactly the Paper 6 tensorial closure rather than an alternative scalar-tensor or higher-derivative geometry.

**Gate 24.1** (Tensorial geometric response gate). *The scalar stiffness  $K_t$  must lift to a tensorial comparison-cost structure  $g_{ab}$  or equivalent cost kernel such that matter recruitment load changes the full comparison geometry, not only the static time-comparison sector.*

**Gate 24.2** (Self-coupling gate). *The geometric recruitment load must include the cost of the geometric substrate's own recruited configuration. In GR language, gravitational field energy gravitates. In cost language, committed comparison-cost distortion contributes to the future comparison-cost kernel.*

**Gate 24.3** (Einstein-equation gate). *The nonlinear geometric recruitment functional must have Euler-Lagrange equations equivalent, in the represented smooth Lorentzian regime, to*

$$G_{ab} + \Lambda g_{ab} = \frac{8\pi G}{c^4} T_{ab},$$

with  $G$  remaining either a substrate-specific input or a derived stiffness constant.

*Boundary 24.4* (Primary translation theorem). The Paper 6 endpoint can be restated in Paper 9 language as follows:

*If finite comparison continuations admit a complete calibrated response geometry, a physical comparison-response anchor, a smooth local Lorentzian signed comparison kernel, universal recruitment coupling, local A9-type closure, and nonlinear self-consistent response, then their represented continuum equations are the Einstein equations.*

Paper 6 proves the represented-geometry closure under its hypotheses. Paper 9 has not yet proved that finite continuability forces those hypotheses. Its current hard local target is the weak-field trace theorem  $\theta = 0$ , the scalar shadow of the same closure.

| Object                               | Status          | Role here                                                                                                                  |
|--------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------|
| Held continuation profile            | I               | Imported from Papers 1, 10, and 14 as the pre-record identity carrier.                                                     |
| Record-forming class transition      | I               | Imported from Paper 37; downstream of the held response geometry.                                                          |
| Complete represented profile carrier | I               | Imported from Papers 1 and 10; Paper 5 supplies the quantum-sector completion.                                             |
| Local quantum orientation            | I, quantum only | Imported from Paper 5 only in its declared coherent quantum sector; not identified with clock-ruler or causal orientation. |
| Geometric response quotient          | P H             | Proved here under constant-rank regularity.                                                                                |
| Substrate anchor                     | O/C             | Identity-complete physical anchor still to be constructed.                                                                 |
| Lorentzian signed comparison kernel  | O/C             | Requires independent cone and signed two-homogeneous gates.                                                                |
| No-trace weak-field source           | O/C             | The invariant target remains $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$ .                                              |

## 25. Input ledger and closure plan

| Claim / object                                                      | Status | What remains                                                                                                                     |
|---------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------|
| Continuation judgment<br>$\Gamma \Vdash_C D \rightsquigarrow E$     | P      | Imported from Paper 10 as primitive grammar.                                                                                     |
| Regular response quotient                                           | P H    | Proved locally under constant rank; the physical comparison response model and its rank stratification remain to be constructed. |
| Pullback process gauge<br>and response distance                     | P      | Proved on every regular quotient; smooth/strongly convex Finsler structure requires extra norm regularity.                       |
| Linear Paper 2 specialization                                       | P      | Proved as the flat quotient branch with optimal upper and lower gains.                                                           |
| Physical-ledger calibration                                         | C/P    | The comparison theorem is proved; the actual ledger-rate bounds and response-class completeness are physical inputs.             |
| Residual energy–distance<br>relation                                | P SRR  | Proved locally when the residual Hessian is positive definite.                                                                   |
| Comparison-response anchor                                          | O/C    | Construct $A_x$ and show the anchored germ agrees with direct comparison cost.                                                   |
| Quadratic positive cost<br>metric                                   | P Q    | Proved from the parallelogram certificate; whether the substrate passes the certificate is open.                                 |
| Signed Lorentz–Finsler<br>comparison structure                      | O/C    | Derive the causal cone, orientation, signed two-homogeneous function, and Lorentzian Hessian.                                    |
| Dimension lock-in                                                   | O      | Requires an anchor-aware mode-count theorem, not merely a register-count argument.                                               |
| Point-source $1/r$ profile                                          | C/P    | Proved from an assumed isotropic quadratic exterior kernel.                                                                      |
| Clock response and red-<br>shift                                    | R      | Conditional recovery under load law, calibration, and proper-time dictionary.                                                    |
| Trace–volume identity                                               | P/C    | Proved in the represented clock–ruler comparison plane.                                                                          |
| Gauge-fixed response Hes-<br>sian                                   | O/C    | Derive $\mathcal{K}$ and $\mathcal{J}$ from the recruitment master equation after gauge removal.                                 |
| No-trace operator<br>$P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$ | O/C    | Decisive local theorem; source-selection and constrained fixed-volume are the two principal routes.                              |
| Spatial response $\gamma_C = 1$                                     | C/O    | Follows from the no-trace operator through the trace–volume theorem.                                                             |
| Conditional weak-field ex-<br>terior                                | C/R    | Follows from clock calibration, universal coupling, and no trace.                                                                |

|                         |     |                                                                                                         |
|-------------------------|-----|---------------------------------------------------------------------------------------------------------|
| Full Einstein equations | I/C | Imported from Paper 6 under H1–H3/A9/Lovelock; residual finite-continuability translation remains open. |
|-------------------------|-----|---------------------------------------------------------------------------------------------------------|

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## 26. Formal theorem tree for closure

| ID    | Claim                                                                                                  | Status | Dependency                    |
|-------|--------------------------------------------------------------------------------------------------------|--------|-------------------------------|
| G9.1  | Constant-rank response gives a local quotient chart.                                                   | P H    | Proposition 12.2.             |
| G9.2  | The process norm pulls back to a quotient tangent gauge.                                               | P      | Theorem 12.4.                 |
| G9.3  | Intrinsic response length dominates endpoint process displacement.                                     | P      | Theorem 12.6.                 |
| G9.4  | The finite linear chart is flat and recovers Paper 2 gains.                                            | P      | Theorem 12.7.                 |
| G9.5  | Calibrated physical path cost is bi-Lipschitz to response distance.                                    | P H    | Theorem 13.2.                 |
| G9.6  | Smooth residual energy is one half squared local distance.                                             | P SRR  | Theorem 13.4.                 |
| G9.7  | Response-orbit geometry pulls back to the physical substrate only after the anchor gate closes.        | C      | Gate 14.60 and Theorem 14.61. |
| G9.7a | Operational basis to physical precision-channel realization is a named open bridge.                    | O/C    | Gate 14.36.                   |
| G9.7b | Equal closed-world fibres need smooth response-factor completeness for the derivative-kernel identity. | C      | Gate 14.53 and Theorem 14.54. |
| G9.7c | The response-orbit quotient gauge is intrinsic; a physical site anchor is a separate gate.             | P/O    | Theorem 14.58 and Gate 14.60. |
| G9.8  | Parallelogram quadraticity gives a positive cost metric.                                               | P Q    | Proposition 14.62.            |
| G9.9  | Positive pullback geometry cannot supply Lorentzian signature.                                         | P      | Proposition 15.1.             |
| G9.10 | A signed Lorentz–Finsler comparison structure represents the causal cone.                              | O/C    | Gate 15.4.                    |

|       |                                                                                         |     |                                                 |
|-------|-----------------------------------------------------------------------------------------|-----|-------------------------------------------------|
| G9.11 | Isotropic quadratic exterior recruitment gives $\chi \sim 1/r$ .                        | C/P | Proposition 16.4.                               |
| G9.12 | Calibrated clock response recovers redshift.                                            | R   | Response law and proper-time dictionary.        |
| G9.13 | Generator trace equals comparison-volume variation.                                     | P/C | Theorem 21.33.                                  |
| G9.14 | Linear loaded equilibrium is $h_\chi = \chi \mathcal{K}^{-1} \mathcal{J} + O(\chi^2)$ . | P H | Theorem 21.23.                                  |
| G9.15 | The no-trace gate is $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ .                 | O/C | Source-selection or constrained-volume theorem. |
| G9.16 | No trace gives $\gamma_C = 1$ and the first-PN exterior.                                | C/R | Corollary 21.34 and Theorem 21.49.              |
| G9.17 | The represented nonlinear closure is Einstein gravity.                                  | I/C | Paper 6 H1–H3/A9/Lovelock import.               |

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## 27. Minimal publishable core

A stable geometry-centered version of Paper 9 should contain:

1. the regular response quotient and pullback-gauge theorem;
2. the intrinsic path-distance and linear Paper 2 recovery theorem;
3. the physical-ledger calibration theorem and the length–energy fork;
4. the comparison-response anchor and the explicit separation of positive cost from Lorentzian comparison geometry;
5. the trace–volume theorem in the loaded clock–ruler plane;
6. the no-trace operator reduction  $P_{\text{tr}} \mathcal{K}^{-1} \mathcal{J} = 0$ ;
7. the existing conditional redshift and first-PN recovery package, with Paper 6 imports kept explicit.

This is a stronger mathematical identity for the paper than “cost accounting plus a weak-field bridge.” It makes Paper 9 the intrinsic differential geometry of admissibility response while preserving the precise gravitational target.

*Audit Note 27.1* (Defer connections and curvature). A connection is not canonical on a general nonsmooth or non-strongly-convex Finsler structure, and rank-changing response spaces may not be manifolds. Curvature should therefore be introduced only after the regular quotient, smoothness, strong convexity or quadraticity, anchor, and causal signature gates have closed. The trace–volume theorem does not require that later machinery.

## 28. Research tasks for the next draft

1. **Construct the physical response manifold.** Identify the actual comparison-response coordinates in the recruitment formalism, compute the rank of the response map, and classify rank-changing strata.

2. **Calibrate the ledger rate.** Prove bounds  $aF_{\text{resp}} \leq j \leq bF_{\text{resp}}$  on a complete declared response class, or exhibit fixed setup, threshold, or hysteresis terms that force a non-length branch.
3. **Derive the comparison-response anchor.** Specify how an infinitesimal site displacement recruits a quotient response direction and prove agreement with the direct comparison germ.
4. **Test quadraticity rather than assume it.** Measure or derive parallelogram defects. Retain the Finsler branch if weighted or polyhedral cost survives.
5. **Build the causal comparison function.** Derive the finite-coordination cone, irreversible orientation, and a signed two-homogeneous  $Q^{\text{cmp}}$  with Lorentzian Hessian.
6. **Compute the loaded response operator.** Expand the Papers 24/25 recruitment master equation in the comparison-kernel sector to identify  $\mathcal{K}$  and  $\mathcal{J}$  after gauge removal.
7. **Prove the no-trace operator.** Pursue either  $P_{\text{tr}}\mathcal{J} = 0$  plus Hessian block-diagonality, or a primitive fixed-volume constraint. The target is  $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$ , not merely a fitted  $\gamma_C$ .
8. **Interpret the weight-one program microscopically.** Treat its capacity accounting as a candidate mechanism for source-selection, not as a substitute for the general response theorem.
9. **Freeze Paper 2.** Once these theorems stabilize, Paper 2 should import the finite linear specialization and its physical-ledger bridge rather than develop a parallel geometry section.

## A. Candidate microscopic realization of the no-trace gate: the weight-one program

The trace gate can be priced in capacity terms, and the price splits. Write the static response as a direct clock coupling plus a spread deposit of weight  $w$ , distributed  $\eta$ -democratically over the comparison cell with denominator  $n - 1$ , where  $n$  is the spatial dimension. We adopt this decomposition and this spread shape; neither is derived, and the ledger below carries them as S0 and S1. The spatial coefficient is then a one-parameter family,

$$\gamma_C(w) = \frac{w}{(n - 1) - w},$$

and general relativity is the weight-one point in every dimension:  $\gamma_C(1) = 1/(n - 2) = \gamma_{\text{GR}}(n)$ . The closure candidates for the response sit at fork coordinates on this curve — the plane rule at  $w = 1$ , the four-cell rule at  $w = 1/2$ , the per-channel rule at  $w = 0$ , the pooled rule at no finite weight — so the gate question becomes a question about one scalar weight (`check_T_gammaC_carrier_fork`, `gamma_c_carrier_program.py`).

The weight has an exact accounting reading. Let  $D$  be the cell-sum of a deposit pattern over the  $n + 1$  channel accounts of the comparison cell. Then  $w = D(\text{spread deposit})/D(\text{commitment})$ , identically in  $n$  and  $w$ : the  $\eta$ -diagonal sums to  $n - 1$  over the cell, so the denominator is not an independent import but the unique normalization under which the weight reads as a fraction of the committed amount. The same cell-sum annihilates every clock-ruler exchange pattern, so the redshift sector carries zero net amount and cannot be over-billed by this accounting. And  $w \neq 1$  is exactly one identically conserved scalar improvement mode of strength proportional to  $1 - w$  — the independent scalar comparison mode of Proposition 21.45, met in ledger coordinates (`check_`

T\_weight\_one\_reduction, gamma\_c\_carrier\_program.py). In this coordinate the no-trace gate is a conservation statement: the substrate’s cell-side deposit matches the load’s committed amount.

Matching splits into two halves with different prices. The no-source half needs the minimal trace-currency premise TCP-min — the response deposit is denominated in committed capacity at a channel-uniform rate — a named conjecture, not a theorem. Under it the deposit must be billed to the load’s own committed line: no census sector furnishes a second local payer, and a line cannot be overdrawn. Hence  $D(\text{deposit}) \leq D(\text{commitment})$ , that is  $w \leq 1$  and  $\gamma_C \leq 1/(n-2)$ ; at  $n = 3$ ,

$$\gamma_C \leq 1 \quad \text{at} \quad [P_{\text{structural}} \mid \text{S0/S1} + \text{TCP-min} + \text{H5-W} + \text{census legs} \\ + \text{the Paper 9/Paper 10 base (P5, B7 incl. adopted } C_M)].$$

The bound is citable only with that ledger attached. H5-W is census transport re-fenced to net-deposit objects — a named premise, separate from the rest and still standing on its own. P5 is this paper’s scalar package; B7 is Paper 10’s held-record capacity floor together with the adopted identification of the maintenance cost  $C_M$  with committed capacity. The no-sink half costs more. It needs TCP-strong, the re-expression-in-full clause: the static response is the same committed amount read at the cell, not less — the one-sided clause  $D(\text{deposit}) \geq D(\text{commitment})$ . Nothing cheaper closes it, because an absence of deposit separates no continuation profile and carries no bill. Composed with the bound, TCP-strong gives  $w = 1$ , hence  $\gamma_C = 1/(n-2)$  in every  $n$  and  $\gamma_C = 1$  at  $n = 3$  — still conditional on the full ledger. Neither trace-currency clause is derived.

The TCP-strong gap has a precise location in Paper 10. Testedness there is binary: a distinction-expression either separates a tested continuation profile or preserves every one. The amount-graded refinement that would close the gap — billing a partial expression for its missing fraction — does not derive from the calculus. The obstruction is an encoding wall: the calculus grades the tested-atom count, the count is representation data, and floor-coherent signatures hold the same record at grades  $\varepsilon^* \cdot \{1, \log_2 N, N\}$  depending on how it is atomized, with  $\varepsilon^*$  the distinction floor (check\_L\_amount\_graded\_testedness\_encoding\_no\_go, amount\_graded\_testedness.py). What survives is an anatomy: over the adopted linear frame the discharge splits into a distributed-encoding existence premise and a par billing rate, with the dimension audit selecting par within the linear family — a selection, not a discharge. Whether the substrate holds a static response profile as one compressed register or as one tested unit per resolution rung is a record-structure question about the substrate, and it is the single genuine reopening door.

The door has a shape. In the coherent shared-read model, the only class of activation law that moves the encoding argmin toward the distributed register is a strictly superadditive read-contention law; linear and subadditive laws leave the compressed register in place (check\_T\_fagt2\_encoding\_argmin\_pressure\_and\_read\_channel, amount\_graded\_testedness.py). And the banked corpus prices every compatible-sharing configuration on the shut side of that gate. The derived joint-cost identity  $\Delta = \varepsilon^*(J - R)$ , iterated over  $M$  compatible readers of one shared record channel, prices the sharing at  $\Delta = -(M-1)\varepsilon^*$  — a strictly deepening discount; the banked activation floor is linear and saves nothing; and the recruitment corpus is reader-blind by census, with no demand-multiplicity variable anywhere in its banked signatures. The one sign-positive source in the calculus is irreducible jointness,  $J > R$  — the occupancy feature. So the residual question is an occupancy fork. Either the concurrent-activation bottleneck sits at sub-record granularity — finiteness supplies the superadditive shape of such a cap for free, but the granularity is a premise finiteness does not force — or concurrent readouts of the static record are irreducibly joint at atom granularity: occupancy at the read interface, in the current Paper 5 sense of a non-Boolean held-profile sector with record-incompatible co-available continuations, per-interface and read off the world rather than

derived, and not adopted here. The bill for the second horn is stated plainly: it needs the static-field read interface to present record-incompatible co-available families, and a classical broadcast readout of one static record is the separable configuration par excellence, so the natural typing sits on the adverse side (`check_T_contention_low_granularity_occupancy_fork, amount_graded_testedness.py`).

The bound meets observation signed. Cassini (Bertotti–Iess–Tortora 2003) gives  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ , and the central value sits  $0.91\sigma$  above the bound: the bound survives at under one sigma and is under live one-sided tension, not merely saturated. A confirmed  $\gamma_{\text{obs}} > 1$  at  $\geq 3\sigma$  falsifies the composite (S0/S1 + TCP-min + H5-W + census + P5/B7 base). A factor-three precision improvement at the same central value puts the composite under roughly  $3\sigma$  pressure. A confirmed  $\gamma_{\text{obs}} < 1$  at  $\geq 3\sigma$  refutes only the no-sink clause and leaves the bound standing. The two halves are separately falsifiable, which the unsplit gate was not.

The decisive target remains the no-trace/unimodular certificate of Gate 21.41: by Proposition 21.42 and Corollary 21.43 it closes the leading spatial coefficient, and by Propositions 21.28 and 21.31 any trace source is exactly the PPN- $\gamma$  obstruction. What the reduction contributes is the exact residue: over the adopted spread shape, the gate’s underived content is two currency clauses, and the cheaper one already buys a falsifiable one-sided bound.

## B. Source-of-record redshift calculation

This appendix records the compact calculation that motivates the first recovery theorem. Let a static point mass carry an effective accumulated recruitment load  $\chi_{\text{rec}}(r)$ . Suppose the exterior linear-response law is

$$K_t(r) = K_{t0}(1 - \alpha\chi_{\text{rec}}(r)),$$

and suppose the point-source equilibrium plus physical calibration gives

$$\alpha\chi_{\text{rec}}(r) = \frac{2GM}{rc^2}.$$

Then

$$K_t(r) = K_{t0} \left( 1 - \frac{2GM}{rc^2} \right).$$

Using the represented proper-time dictionary,

$$d\tau(r) = \sqrt{K_t(r)/K_{t0}} dt,$$

one obtains

$$d\tau(r) = \sqrt{1 - \frac{2GM}{rc^2}} dt.$$

For two static observers at  $r_B < r_A$ , equal coordinate periods correspond to different local proper periods, hence

$$\frac{\omega_A}{\omega_B} = \frac{d\tau(r_B)}{d\tau(r_A)} = \sqrt{\frac{1 - 2GM/r_Bc^2}{1 - 2GM/r_Ac^2}}.$$

Expanding to first order gives

$$\frac{\omega_A}{\omega_B} \approx 1 - \frac{GM}{c^2} \left( \frac{1}{r_B} - \frac{1}{r_A} \right).$$

This is the static gravitational redshift form. The calculation is a conditional recovery because the response law, calibration, and proper-time dictionary remain open gates.

## C. Minimal next-lemma inventory

The next version should prove, weaken, or abandon the following lemmas.

1. **Regular-quotient lemma.** The comparison-response map supplied by the recruitment formalism has constant rank on the weak-field branch, or else admits a controlled stratification.
2. **Ledger-calibration lemma.** The physical interaction rate is uniformly comparable with the pullback process gauge on the complete response class.
3. **Anchor lemma.** The direct comparison germ equals the anchored response germ for all locally admissible comparison directions.
4. **Quadraticity-or-Finsler lemma.** Determine whether the local gauge passes the parallelogram certificate; if not, retain the exact Finsler geometry rather than fitting a metric.
5. **Causal-cone lemma.** Finite coordination and irreversible record order produce a smooth oriented cone and signed comparison function with Lorentzian Hessian.
6. **Response-Hessian lemma.** Loaded-equilibrium expansion yields a gauge-fixed invertible  $\mathcal{K}$  and source  $\mathcal{J}$  in the comparison-kernel sector.
7. **No-trace source lemma.** Prove  $P_{\text{tr}}\mathcal{J} = 0$  and  $P_{\text{tr}}\mathcal{K}P_{\text{TF}} = 0$ , or derive an admissibility constraint enforcing  $\text{tr}_{q_0} h = 0$ .
8. **Self-coupling lemma.** The cost of geometric distortion contributes to later recruitment load and lifts the weak-field response into the Paper 6 nonlinear closure.
9. **Ledger-divergence lemma.** Relabeling comparison sites induces an identically conserved geometric ledger compatible with matter continuation.

## D. Closure certificate

This supplement is dependency-closed as a conditional proof package when every open gate above appears as a typed node in the theorem ledger. Dependency closure does not mean physical realization closure. In particular, finite joint extension on arbitrary operational closures, the faithful operational-to-channel realization, finite smooth generation, smooth response-factor completeness, the physical substrate anchor, quadraticity, signed causal geometry, the physical no-trace source law, and the nonlinear Paper 6 endpoint remain conditional, open, or external as explicitly marked.

## E. Conclusion

Paper 9 is now organized around a precise geometric claim. Continuation is primitive, while geometry is represented continuation. A structurally aligned held continuation profile is first represented in a complete carrier and process algebra. A finite response model then loses its redundant implementation coordinates by quotienting response fibers. The represented process norm then pulls back to a positive tangent gauge, whose path integral gives intrinsic response distance. In a linear chart this geometry is flat and reproduces the quotient norm, optimal upper gain, positive minimum gain, and compatible bi-Lipschitz comparison already established in Paper 2. A separately



calibrated physical ledger may equal that distance, remain only comparable with it, or fail to be a path metric because of setup, threshold, indivisibility, saturation, or hysteresis.

The same architecture resolves the homogeneity ambiguity. One-homogeneous path ledgers generate length. Smooth residual penalties generate local quadratic energy, and their correct geometric form is one half squared distance. Neither branch should be renamed into the other. The comparison-response anchor is now placed on a derived record differential space. Closed-world precision channels generate the topology; lossless postprocessing preserves the physical quotient; smooth saturation supplies the Sikorski functional calculus; finite capacity gives finite-channel sufficiency; finite differential sufficiency identifies the subcartesian locus; and locally complete admissible continuation derivations generate immersed smooth manifold orbits. The anchor therefore acts on emergent smooth continuation branches rather than on an assumed substrate manifold. Under quadraticity it yields a positive cost metric, but a separate causal-cone and signed two-homogeneous gate is required for Lorentzian comparison geometry. This blocks the misleading shortcut  $g = T^*T$  as a derivation of spacetime signature while preserving the valid pullback construction for positive cost.

The gravitational application becomes sharper rather than weaker. Recruitment load deforms the represented clock-ruler comparison kernel. The generator trace is exactly the first variation of the local comparison-volume element, and clock calibration turns that scalar volume mode into the PPN obstruction  $1 - \gamma_C$ . In the gauge-fixed loaded-equilibrium problem the entire no-trace question is

$$P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0.$$

This equation separates two routes: show the scalar source is trace-free and the Hessian does not mix trace sectors, or derive a primitive fixed-volume admissibility constraint. The weight-one capacity program may supply a microscopic source-selection mechanism, but the invariant operator equation is the theorem it must discharge.

Paper 6 remains the represented smooth source of record for the Einstein closure. Paper 9 supplies the intrinsic response geometry underneath it: quotient, pullback gauge, path distance, ledger calibration, anchor, causal signature gate, recruitment deformation, and trace-volume response. That is the coherent role of the paper and the correct home of the interaction-geometry program.

**Compressed result.**

Held continuation profiles supply the pre-record identity carrier. Response-null quotienting of their represented process realizations produces intrinsic process-induced geometry. Physical cost equals that geometry only after ledger calibration. The comparison substrate inherits it only through an anchor. Quadraticity yields a positive cost metric; causal signature requires a separate signed gate. Under recruitment load, the scalar weak-field obstruction is the local comparison-volume mode

$$\text{tr } L = \frac{d}{d\chi} \log \sqrt{|\det q_\chi|} \Big|_0 = 1 - \gamma_C,$$

and its decisive microscopic target is  $P_{\text{tr}}\mathcal{K}^{-1}\mathcal{J} = 0$ .